

On the Accuracy of Network Telescope Traffic

Alexander Männel

Born on: 8th December 1999 in Chemnitz

Course: Master Informatik (PO 2010)

Matriculation number: 4803563

Matriculation year: 2021

Master's Thesis

to achieve the academic degree

Master of Science (M.Sc.)

Supervisor

M.Sc. Jonas Mücke

Supervising professor

Prof. Dr. rer. nat. Matthias Wählisch

Submitted on: 18th August 2024

Statement of authorship

I hereby certify that I have authored this document entitled *On the Accuracy of Network Telescope Traffic* independently and without undue assistance from third parties. No other than the resources and references indicated in this document have been used. I have marked both literal and accordingly adopted quotations as such. There were no additional persons involved in the intellectual preparation of the present document. I am aware that violations of this declaration may lead to subsequent withdrawal of the academic degree.

Dresden, 18th August 2024

A. 

Alexander Männel

Contents

List of Figures	VI
List of Tables	VII
1 Introduction	1
2 Technical Background and Related Work	3
2.1 Internet Communication	3
2.1.1 Internet Protocol	3
2.1.2 Border Gateway Protocol	3
2.1.3 Issues with BGP	4
2.1.4 Structure of the Modern Internet	4
2.2 Unsolicited Internet Traffic	5
2.2.1 Scanners	6
2.2.2 Backscatter	6
2.3 Network Telescopes	6
2.3.1 Concept	7
2.3.2 Observable Malicious Activities	7
2.3.3 Other Observable Phenomenons Captured by the Telescope	8
2.3.4 Known Limitations	9
2.3.5 Alternative Deployments of Network Telescopes	9
2.4 Packet Loss in Transit	10
2.4.1 BGP Hijacks	10
2.4.2 Middleboxes	11
2.4.3 General Packet Loss	12
3 Measurement Design	13
3.1 Traffic Collection at Network Borders	13
3.1.1 Amount of Traffic Logged	13
3.1.2 Potential Bias of Vantage Points	14
3.2 Active Probing	15
3.2.1 Type of Measurements	15
3.2.2 Third Party Measurement Projects	16
3.2.3 Self Initiated Measurements	17
3.2.4 Summary	17

4	Measurement Setup	18
4.1	Traffic Collection	18
4.1.1	CAIDA Network Telescope	18
4.1.2	Types of Data	19
4.2	Traffic Sources	19
4.2.1	Traffic Collection at Network Borders	19
4.2.2	Known External Scanner Activity	20
4.2.3	RIPE Atlas Probes	20
4.3	Data Processing	22
4.3.1	Extraction	22
4.3.2	Matching Packets from Vantage Points	24
4.3.3	Detecting Packet Loss in Scanners	25
4.3.4	Detecting Packet Loss in RIPE Atlas Scans	25
4.3.5	Data Analysis	25
5	Operational Challenges of Network Telescopes	27
5.1	Inaccuracies of the Filterlist	27
5.1.1	Understanding the Used Address Space	27
5.1.2	Filterlist Implementation	28
5.1.3	Anomalies in Observations	29
5.1.4	Identification of Network Telescope Addresses	29
5.1.5	Impact of a Strict Filterlist	32
5.1.6	Impact of a Loose Filterlist	32
5.1.7	HAMnet Data Collection	33
5.2	Network Telescope Under High Loads	35
5.2.1	Broken Captures	35
5.2.2	Dropping Packets	36
6	Results	39
6.1	Connectivity-Related	39
6.1.1	Visibility of Chinatel Customers	39
6.1.2	Impact of Hops to the Network Telescope	40
6.2	Filtering-Related	41
6.2.1	IP Fragmentation Losses	41
6.2.2	Bogon Source IP Filtering	42
6.2.3	UCSD Campus Network Filtering	42
6.2.4	On-Path Filtering	43
6.3	General Packet Loss	44
6.3.1	Low Packet Loss in Tier-One Networks	44
6.3.2	Packet Loss due to Link Overload	44
7	Discussion	47
8	Conclusion and Future Work	51
	Bibliography	i

List of Figures

2.1	CIDR notation for network 192.168.178.0 with subnet mask 255.255.255.0	4
2.2	Example for an AS PATH via two Tier-One providers	5
2.3	Scanners send packets to many addresses, including NT addresses.	7
2.4	Network telescope being part of the hosts receiving responses from the victim (“backscatter traffic”)	8
2.5	Origins of packet loss along the packets path. In this work, losses after the source are considered.	11
3.1	Example path of a packet with different possible locations for packet loss (lightning). Transit Provider #1 is the VP, so the path between it and the NT is observed (blue arrow).	14
3.2	Visualization of a traceroute arriving at the NT	15
4.1	CAIDA Network telescope capturing architecture [7]	18
4.2	AS distribution of probe set as reported by RIPE Atlas. The size of each AS represents the number of probes participating from that network.	21
4.3	Country distribution of probe set as reported by RIPE Atlas. The size of each country represents the number of probes participating from that nation.	21
4.4	Data processing setup to search for lost packets.	23
5.1	Hilbert curve of AMPRnet, IP ranges colored in blue are granted leases.	28
5.2	Hilbert curve of AMPRnet, where DTAs are colored red, ATAs are marked black and subnets from the AMPRnet list blue.	30
5.3	Hilbert curve of AMPRnet, with uDTAs covered by more specific BGP announcements colored yellow.	31
5.4	Hilbert curve of AMPRnet, with addresses covered by more specific BGP announcements colored yellow and entries of the subnet list expanded to their covering /24.	31
5.5	Number of packets received from SC2. The green line shows the introduction of the new filterlist.	33
5.6	Hilbert curve of AMPRnet, with a highlighted area where only some IP addresses within a block received scanner traffic.	34
5.7	Size of stored PCAPs files of NT and bandwidth utilization.	35
5.8	Measurements of SC2 with packet drop in relation to total packets received by NT. . .	36
5.9	Measurements of SC2 with packet drop in relation to total packets received by NT on Nov 22nd	37
5.10	Resource Utilization of NT capture machine on Nov 22nd	37

List of Figures

6.1	Number of packets received by each source of SC2 and the top 90% of TTLs by each source.	40
6.2	Number of packets received by each source of SC2 and the average TTL by each source.	41
6.3	Different combinations of measurements for which probes were able to reach the NT.	43
6.4	Packet loss for each source of SC2 during the observation period.	44
6.5	Packet loss event of PCCW source of SC2 from November 12th to 15th 2023.	45
6.6	Packet loss event of Hurricane source of SC2 on December 4th 2023.	45

List of Tables

4.1	Overview of Vantage Point sources	20
4.2	Overview of Scanner sources	20
4.3	Tags used in the selection of the RIPE Atlas probes	22
4.4	Scan regime for RIPE Atlas probes	23
5.1	IP Protocol distribution of average IBR (CAIDA NT wide statistic) and the observed misconfigured HAMnet Source.	34
6.1	Packet loss from RIPE Atlas probes for: normal pings, first fragmented packet, second fragmented packet	41
6.2	Filterlist of UCSD campus network	42

1 Introduction

Network telescopes have been a vital tool for the Internet measurement research community for a long time. They enable the analysis of Internet Background Radiation (IBR) by listening on IP addresses without any services hosted on them. Traffic collected by network telescopes has been used extensively for the research of outages [5, 10, 22], Internet censorship [11], DDoS attacks [14, 27, 46] and Internet scanners [18, 25, 32, 51]. Conducting studies using longitudinal telescope data allows for the observation of changes and trends of DDoS attacks and scanner activity. With cyberattacks becoming more frequent, the interest in understanding them and the entire ecosystem grows. As network telescopes offer a unique view on the mainly malicious IBR, data collected by them will remain highly relevant.

The reliance on network telescope requires them to accurately represent the IBR received by other hosts connected to the Internet. If specific types of packets are lost towards a telescope compared to other networks, the telescope is unrepresentative. Being unreachable for some topological or geographical region can result in a skewed view on IBR, as traffic from sources unable to reach the telescope is excluded. Such blind spots have to be considered by work utilizing telescope data, otherwise conclusions may be drawn too broadly or even be incorrect. Additionally, any assumptions made on the size of the network telescope need to be true. To scope the volume of IBR present in the Internet, knowing the relative size of the telescope in comparison to the rest of the Internet is essential. Getting this relative size wrong will result in over or under claiming the relevance of observed phenomena and activities.

For a network telescope to fulfill its role, it needs to record any packet sent to it. For this, the first fundamental challenge is to have a functioning path for the packet to take. Devices between the source such as routers and middleboxes could for example filter packets they deem unsafe. A packet can also fall victim to the general packet loss within the Internet, or be impacted by a disruption to a network operator. The second challenge arises once the packet has been received by the telescope. Recording and storing the large amount of traffic involves complex setups, which can suffer issues and outages. Issues get amplified by the ever increasing amount of traffic sent in the Internet. Operating and monitoring a network telescope and its adjacent infrastructure is complex, while at the same time any small disruption has a direct negative impact on the quality of the packets recorded.

In this work, we aim to quantify the packet loss both towards the telescope and in the recorded data. To answer this question, we use active and passive measurements to establish a ground truth expected to arrive at the CAIDA network telescope located at the University of California San Diego. By comparing the traffic recorded by the telescope with the expected traffic, unique and precise insights into the telescopes blind spots and issues are gained. The findings of this work can be divided into **On-Path Challenges**, where packet loss occurs on the path towards the telescope and **Operational Challenges**, directly involving the telescope infrastructure.

On-Path Challenges:

1. We showcase an incident involving a large Chinese ISP, where BGP misconfiguration results in the unreachability of the network telescope for all the ISPs customers. The incident shows that such cases exist, but have not been considered before.
2. We analyze traffic collected at an IXP and a transit provider. The analysis shows that general packet loss between transit providers and the network telescope is below 0.05%, but incidents of individual providers can cause spikes. We also observe that no packets with bogon source addresses observed by the transit provider reach the telescope.
3. We perform active traceroute measurements using RIPE Atlas to search for on-path filtering close to the source devices. We find that filtering based on protocol and port happens, with dropping of TCP being the most prevalent (3.6%). 93% of probes are able to reach the telescope, showing that filtering is not widespread. Additionally, by sending packets larger than 1500 Bytes, we can observe a seven fold increase in packet loss for fragmented packets.

Operational Challenges:

1. We provide a comprehensive view on the current state of the address space of the CAIDA network telescope. It shows that the network telescope is significantly smaller than assumed by some related work.
2. We show that the current process implemented to filter out non-IBR traffic is not adequate. Due to filtering on a /24 prefix level, the effective size of the telescope is decreased. On the other hand, major active areas spanning at least two /16 prefixes were not excluded, leading to at least six months of non-IBR traffic stored. This traffic has drastically different properties compared to normal IBR.
3. We provide concrete evidence for the impact of changes to the filterlist. This breaks assumptions by related work, as such large changes go unnoticed when interacting just with the telescope data. In our observed case, the size of the telescope decreased by 6.3% in a single update.
4. We show that the telescope experiences partial or complete packet loss under high load. While complete loss is well detectable, the partial loss is not visible through any ordinary monitoring metrics. Under high packet loads over 3.5 million packets per second, the telescope loses 10% of total traffic received, possibly more.

To date, there has been no comprehensive research into the reliability of telescope data. While much research with the help of network telescopes has been conducted [5, 10, 11, 14, 18, 22, 25, 27, 32, 46, 51], no methodological investigations have been performed. Related work considering the exposure and quality of network telescope data does so in comparison to their newly proposed telescope architectures [18, 38, 50]. While such comparisons are useful to evaluate these new concepts, it does not allow for a general conclusion about the accuracy of data collected by network telescopes. To achieve more insight into the accuracy, this work used ground source traffic collected at different observation points as well as through active measurements. We are able to uncover limitations that are not visible in related work, if both the old and new telescope concept are similarly susceptible to it, for example. We provide an unprecedented evaluation of network telescopes as measurement instruments.

This work is structured as follows: A brief overview over the basics of Internet Routing, network telescopes and potential phenomena observed by them is provided in Chapter 2. The Chapters 3 and 4 explain the measurements and data sources used in this work. Chapter 5 presents challenges faced when working with the CAIDA network telescope. Finally, Chapter 6 showcases the results obtained through different measurements and observations. These results are discussed in Chapter 7 and a conclusion and outlook is given in Chapter 8.

2 Technical Background and Related Work

2.1 Internet Communication

End-to-end communication over the Internet is handled by a multitude of protocols working together. Data exchanged between two systems gets sliced up into individual packets. The handling of each packet is performed by protocols categorized by the OSI model, one of which being the Internet Protocol (IP). To allow for routing between the different networks making up the Internet, additional protocols such as the Border Gateway Protocol (BGP) are utilized.

2.1.1 Internet Protocol

IP is the predominantly used protocol for the network layer within the Internet. It is connectionless and just handles the delivery of single packets from the sender to one or more targets. This is in contrast to for example Publish-Subscribe architectures, where a target can decide whether it wants to receive data or not. To allow for connections and more complex interactions, protocols in upper layers of the OSI model can be used.

Addressing within IP is done via IP addresses. For IPv4 these have a length of 32 Bits, while the newer version IPv6 has increased the address length to 128 Bits. A main reason for the larger address space in IPv6 is the increased number of connected devices. The smaller IPv4 address space is already exhausted [15] and the prices for a single IPv4 address have risen significantly [21] in the last years.

For both versions of IP, all possible addresses are known. Due to the complete utilization of the IPv4 address space and its size however, its very easy to iterate all actively connected hosts by pure exhaustive search. In this work we will only interact with IPv4, since the systems we will be using do not support IPv6. For finding active hosts in IPv6 there are some different challenges [16], which will not be covered here.

2.1.2 Border Gateway Protocol

The Internet is made up of a lot of independently managed networks. To allow routing between these different domains the Border Gateway Protocol (BGP) is used.

Each of these networks is called an “Autonomous System” or AS. Every AS is identified by an AS number. These numbers are allocated by the “Internet Assigned Numbers Association” (IANA) in pools and assigned by a delegate Regional Internet Registry (RIR). These RIRs also handle the allocation of IP address blocks to their registered Autonomous Systems.

Each of these networks has a set of IP addresses that it manages and provides connectivity to. This set of IP addresses is made up of a list of IP address blocks which are commonly represented in

“Classless Inter-Domain Routing” (CIDR) notation (see Figure 2.1). Since the number at the back denotes the length of the subnet mask, it also indirectly denotes the number of IP addresses within the address block.

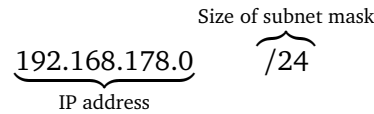


Figure 2.1 CIDR notation for network 192.168.178.0 with subnet mask 255.255.255.0

Not every AS participating in the Internet is directly connected to every other AS. Generally, an AS will have a limited number of partners it connects to, called “peers”. It is to these peers, that every AS “announces” an ASPATH containing only its own AS number to the address blocks that it provides connectivity to. Its peers can now route traffic to these IP addresses originating from their own network. An AS can also choose to advertise a known path to its peers. Contained in such an advertisement is a longer ASPATH, which denotes the intermediate networks en route to the target. These path announcements make it possible for Autonomous Systems to reach networks they are not directly peering with.

To allow for truly global routing, ASes can propagate path announcements to their peers.

2.1.3 Issues with BGP

The current version of BGP was created in 1994 [42]. Back then, not a lot of focus was placed on security. Since every peer has to be set up in the edge routers manually, a certain level of trust between peers was assumed. Today, we know that BGP is very susceptible to both malicious actors and misconfigurations.

With every AS being a trusted entity in the eyes of BGP, no validation on the paths or addresses a peer announces is performed. This allows for so called “BGP hijacks” or “Route leaks”, where a (either intentionally or unintentionally) misbehaving AS announces routes that do not exist or should not be published. If this announcement is propagated by its peers and it gets selected as the preferred path, traffic for these IP addresses will be diverted to the rogue AS. In its simplest form, this is a denial-of-service attack. It also allows for more complex man-in-the-middle attacks, if the traffic is forwarded to the original target afterwards.

BGP hijacks in the wild happen for a variety of reasons, ranging from misconfigurations [45] and scams [41] to censorship [35] and surveillance [12].

The introduction of features like resource public key infrastructure helped combat BGP hijacks. However, lack of adoption or sometimes even the impossibility to adopt such measures means that these are not solved issues. While best practices are in place to protect against route leaks, they still occur reasonably often. BGP is a very good example for how hard it is to update or significantly change core components of the Internet.

2.1.4 Structure of the Modern Internet

Another thing not envisioned by the original draft of BGP was the number of ASes. There are currently well over 100 000 ASes registered. Within this big collection different types of ASes exist. Many Internet Service Providers mostly supply connectivity to their end users. As a result their networks have high inbound traffic. Other network operators, such as hosting or service providers mostly accustom nodes that provide some sort of data. Their networks are typically outbound heavy.

While it would be beneficial for these ASes to peer with one another, it is not always the case. The main reason for this is cost. In order to peer with a partner, both networks need to have at least a point of presence (PoP) in common, where they can physically connect and share traffic. Internet

exchange points (IXP) provide such common locations, but geographic distance or the costs associated with joining an IXP still complicate direct peering.

For this reason, there is another type of network, which we will call “Transit Providers”. These companies maintain networks with many points of presence. Depending on their scope, they may have regional or global reach. If two ASes want to reach each other but do not share a common PoP, chances are high both share one with a transit provider. Via this network, both ASes can now reach each other. The involved infrastructure often includes sea cables and dedicated fiber lines. Since running these global backbones is expensive, transit providers charge both parties for the supplied connectivity. In contrast to the mutual peering talked about before, where both peers exchange traffic free of charge, this creates a customer-provider relationship. These are not directly visible in BGP but can often be inferred.

Almost all ASes have at least one upstream provider they pay for connectivity. A small fraction can access all networks without paying for traffic. They are often called “Tier-One Providers” and are made up of the biggest transit ASes. By having mutual peering between one another, they can reach every AS connected to the Internet. A Tier-One will however only mutually peer with networks of similar global reach and balanced traffic between them, so the number of them stays relatively small [36].

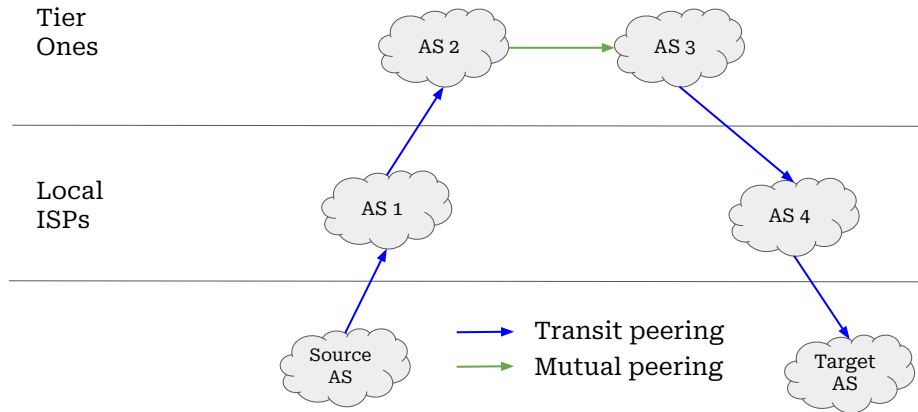


Figure 2.2 Example for an AS PATH via two Tier-One providers

The existence of Tier-Ones has centralized the Internet. A lot of global traffic flows through at least one such provider on the way to its destination. Some of them handle more than 100 Tbit/s [40]. While a network will always prefer its mutual peers instead of its upstream provider, often paths look like the example in Figure 2.2. The source AS wants to route a packet to the target AS. The only path it knows is via its upstream provider AS 1. It in turn routes the packet to its upstream, in this case a Tier-One. Note that in other cases there might be more layers involved and they do not have to be the same number on the side of the receiver. The Tier-One now forwards the packet to a mutual peer which provides connectivity to AS 4. That network is the upstream provider for the target AS and therefore hands the packet over. It is possible that paths do not contain a transit network.

2.2 Unsolicited Internet Traffic

Any node in the Internet can receive unsolicited Internet traffic. This is due to the fact, that the address space is known and no connection has to be established to send a packet. There are two common reasons for this unsolicited traffic otherwise known as “Internet Background Radiation” (IBR).

2.2.1 Scanners

Scanners are a type of source for IBR that search for reachable hosts connected to the Internet. This is feasible in IPv4, due to the limited amount of addresses. While there are currently 3.7 billion allocated IPv4 addresses, modern scanning tools can perform simple scans in 45 minutes on consumer hardware [13]. This type of IBR makes up a significant chunk of unsolicited Internet traffic currently observed. As much as 80% of IPv4 packets in recorded IBR contained indicators associated with scanner software in 2021 [18].

There exist both benign and malicious scanners, who are attempting to discover and identify reachable applications. Benign scanners use the collected data to provide information to businesses about their exposure and attack surfaces. They also allow for the assessment of observed IP addresses and help detect dangerous activities by providing indicators of compromise. Researchers also use scans to uncover unpatched or misconfigured devices [34] and their distribution.

Malicious actors however generally do not stop at the discovery step. Scan results are often followed up by exploit payloads for known vulnerabilities [18] or requests looking for unsecured resources [52]. Other reasons for scanning might be the search for new systems that can be abused as amplifiers [25]. These then get leveraged for amplified DDoS attacks.

Scanner traffic is not confined to a single protocol. Depending on the specific port and the expected application behind it, scanners may use TCP or UDP. For general host discovery ICMP can be used as well. While TCP scanning traffic is often small, since a TCP session is always initialized with a TCP SYN packet, this is not necessarily the case for UDP. In order to receive an answer, a request might have to be well formed [48].

2.2.2 Backscatter

Backscatter is a type of traffic that appears when a victim responds to packets with a spoofed source IP address. Such a packet can be sent by a malicious client whose ISP does not implement “Source Address Validation” (SAV). With SAV packets with source addresses not belonging to the IP blocks owned by the providers AS would be dropped by the edge routers. If that is not the case, a host in that network can send packets with arbitrarily set source IP addresses.

Many big network operators have implemented SAV but there is still a considerable amount of unsecured networks [47]. By looking at the AS a spoofed packet was sent from, its possible to narrow down the potential sources, but attribution to a host is generally not possible. Spoofed traffic is often used for circumventing IP based firewalls and blocking measures. A second, potentially more devastating use case is for reflective amplified DDoS attacks. Here, an attacker does not send packets to the victim directly but via an amplifier service. This host responds to the packets, often with much larger payloads than in the original request. Since the source address is spoofed, the response is not sent to the attacker but to the victim. With that, the attacker has increased the attack volume substantially. In practice amplification factors of above 50 000 have been achieved [9].

The responses to these spoofed packets then end up at the host who has the spoofed IP address assigned to it. Since that receiver can not match that traffic to any of its own activities it is usually discarded. Backscatter can be indicative of the amount of spoofing and DDoS attacks happening. Depending on how the spoofer chooses the source addresses set in the packets, specific hosts might see more or less backscatter traffic for their IP addresses.

2.3 Network Telescopes

Network telescopes (NT) are an instrument to capture Internet Background Radiation.

2.3.1 Concept

The basic idea of a network telescope is to listen to and store all traffic that is sent to a block of otherwise unused IP addresses. For this, the address block has to be routed and an infrastructure set up to record all incoming traffic. Since no legit services are addressable via the used address block, the only traffic received should be IBR. A network telescope is therefore a prime instrument to collect scanner activity and backscatter.

Acquiring a suitable address block is difficult. In order to study IBR by looking at captured traffic, no regular traffic should arrive at the addresses. While IBR has some characteristics, it is not possible to safely determine whether a packet is unsolicited Internet traffic or genuine. IP addresses that are used by network telescopes therefore need to be “dark”. There should be no real traffic going towards the address. As Pauley et al. show, the IP addresses also have to be dark for a considerable amount of time in order to not receive any traffic destined for a service hosted at the address before [39]. If caches and other applications are not yet aware of an address change, a potentially large amount of non-IBR traffic can be falsely stored. Such packets then skew the data derived from the telescope. For example, if distribution of destination ports is analyzed but packets from an old service are stored, that port might be incorrectly over represented.

Acquiring a small amount of suitable addresses is also not enough. A larger number of addresses allows for more captured traffic. This in turn makes for a higher probability that interesting events and patterns can be observed. It also increases the confidence that observations are not just small artifacts. This is especially true, if you want to detect DDoS attacks via backscatter, as only a higher volume of traffic from a single target allows for good detection rates. Bigger telescopes are more representative.

2.3.2 Observable Malicious Activities

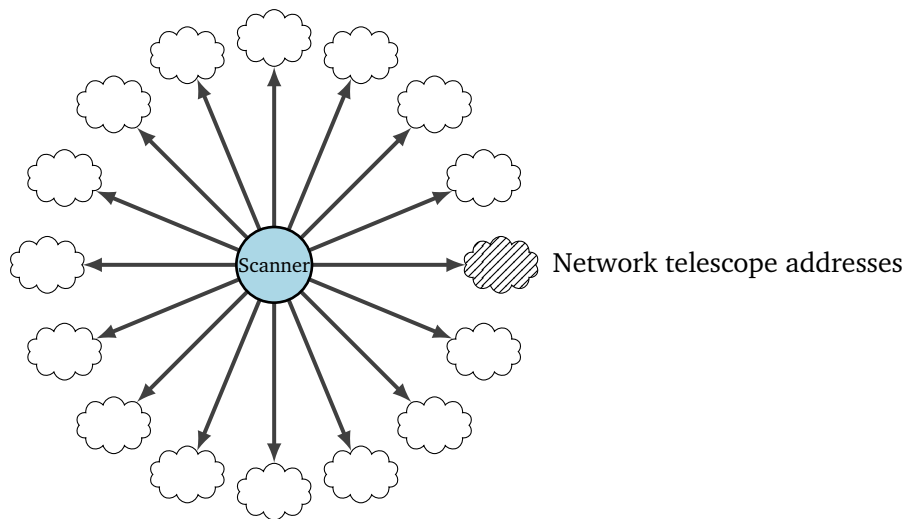


Figure 2.3 Scanners send packets to many addresses, including NT addresses.

Both scanners and backscatter are visible in the traffic captured by a network telescope. By continuously monitoring scanner activity, it is not only possible to generally assess their behavior and goals, but also allows for noticing changes in their behavior. With scanners sending packets to all addresses in the Internet, the telescope also gets scanned, as shown in Figure 2.3.

Besides the general scanners which monitor the Internet for new devices or changes, which are active all the time, there are also opportunistic scanners. These larger, more concentrated scan campaigns often happen when new critical vulnerabilities get discovered. In the wake of the discovery of a remote code execution vulnerability in the “Log4J” library for example, there was a visible uptick in scanners explicitly trying out payloads which could trigger the vulnerability [19].

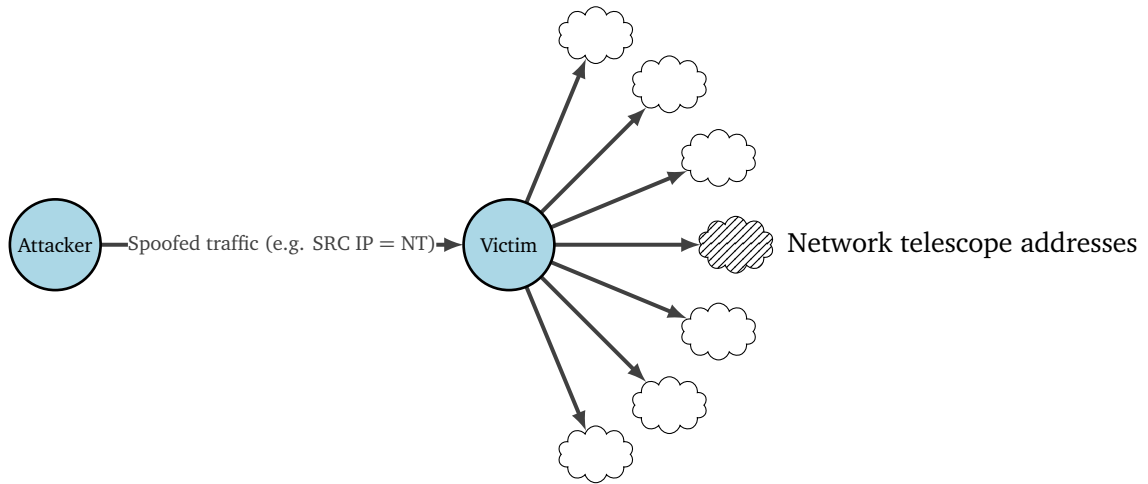


Figure 2.4 Network telescope being part of the hosts receiving responses from the victim (“backscatter traffic”)

By attributing scanning activity to certain campaigns or originating tools, network telescopes also allow researchers to observe the distribution and behavior of worms and other malware. An example for this is the “Mirai”-botnet, which scans specific TCP ports associated with the TELNET protocol. The IBR captured at a network telescope that matches these indicators can provide valuable insights into source devices. Antonakakis et al. showed that infected hosts in the Mirai-botnet are generally IoT- and embedded devices [2]. Leveraging captured traffic, they could observe the spread of different strains in real time and approximate the number of infected hosts.

Given a large enough IP address block, network telescopes are able to observe backscatter from DDoS attacks employing source address spoofing, as depicted in Figure 2.4. A big block is required to capture enough packets to confidently attribute them to an event instead of noise. The collected data helps researchers find patterns in attacker behavior and observe their employed strategies and protocols. By creating a broad overview over DDoS attacks, trends in target selection can be made out. A lower bound for the intensity of an attack can be provided, depending on the distribution of the spoofed source addresses.

Even for amplified reflected DDoS attacks, which generally do not create backscatter, the data from a network telescope can be valuable. The IBR captured contains the scanning effort for vulnerable amplifiers as already mentioned above. As shown by Fachkha, Bou-Harb, and Debbabi, some of these DRDoS attacks do not utilize a previously scanned for set of amplifiers but instead send spoofed packets to every IP address in the hopes of hitting an amplifier [14]. In the paper, they found evidence that the largest DRDoS attack seen at that time used such a technique. Since these packets are easily recognizable in the traffic captured by the network telescope, NTs are valuable even as attackers change techniques.

2.3.3 Other Observable Phenomenons Captured by the Telescope

Thanks to the global distribution of scanners and the relatively constant influx of packets, a network telescope can also observe changes in connectivity of certain ASes. If a region or network loses their connectivity due to a configuration change or an unforeseen event, no packets can reach the telescope. That means that all scanners from the impacted ASes are no longer visible in IBR traffic. By detecting deviations from average scanner activity observed in the past, it is possible to accurately detect AS-level outages [5]. The same work also suggests that it is possible to detect congestion and route changes by comparing historical TTLs of scanner packets with current TTLs, although TTL values can vary for a variety of reasons. This data, combined with other measurements such as active scans, powers the “Internet Outage Detection and Analysis”(IODA) project [22]. It provides

continuous and real time updates on AS-level Internet outages and disruptions. If there were errors in the data used by IODA, it could lead to networks being incorrectly labeled as disrupted.

In some cases, missing scanner traffic can also identify natural disasters which cause connectivity issues. Following an earth quake, changes in the number of observed source IP addresses in the region suggested a loss of Internet access for the affected areas. By geolocating the absent source IP addresses and mapping them in regards to distance from the epicenter, it was possible to even map affected cities [10].

2.3.4 Known Limitations

With the increasing pressure on IPv4 address supply, the costs of creating a network telescope grow as well. With the additional requirement of the addresses being dark, this issue gets amplified as cost saving colocation with other services is not possible. Even existing projects with large IP ranges come under pressure to reduce their holding as they rise in value. In practice, this means a trade off between the size of the address block and operating costs has to be struck. These issues only affect IPv4 network telescopes, as the IPv6 address space is much larger and not under such high pressure. With IPv6 however, other challenges arise, since scanners cannot easily contact every existing IPv6 address. Since the network telescope used in this work only captures IPv4 traffic, this will not be expanded upon here.

Another issue of a network telescope deployed at a fixed IPv4 block is that IBR is location dependent. Often, research evaluates data from multiple network telescopes deployed at different IPv4 address blocks [18, 38, 50]. When these network telescopes have disjoint geolocations, they observe different scanner traffic. This is most obvious in the distribution of the top scanned ports. Depending on the geographic location, different systems and applications might be deployed. Scanner operators have adapted to this and scan different sets of ports for these regions. For example, Hiesgen et al. found that their EU-based installation received port probes for 1433 and 7547, which were not seen at the US-based network telescope [18]. These ports are specific to certain known vulnerabilities of devices dominantly deployed in Europe. Besides the geolocation associated with the IP block, the physical location of the capturing hardware also matters. As shown by Wan et al., some regions, for example Australia, experience higher average packet loss [51]. A network telescope located close to Australian ISPs will therefore capture more scanner traffic from Australia than a telescope in Europe.

Since the IP block a traditional network telescope uses is static, given enough time, the location will become known. This allows scanners and other malicious actors to exclude the telescope range from their activities and therefore make the data collected unrepresentative. While there is not a lot of research regarding this, other work shows that different/newer telescopes receive more traffic than an established system [38].

2.3.5 Alternative Deployments of Network Telescopes

Different solutions have been proposed to circumvent the issue of owning a large IP address block without any active services contained in it.

One approach covered in related work is move the traffic collection from the destination network to a vantage point on the path of the packets [50]. For their vantage points, the authors use various IXPs and collect sampled traffic passing through them.

In a first step, they introduce a classification to identify dark and “grey” IP address blocks by checking if they see any outbound traffic from these blocks. After performing this analysis, they then consider the traffic sent to these identified address blocks as IBR. In their work, they identify ~350k /24 blocks in over 7k ASes. While the paper tries to correct for some issues such as asymmetric routing mostly by employing a large amount of vantage points, they still do not correctly classify all networks. The paper reports a false positive rate of 0.87% for their chosen parameters, while in later figures this rises to up to 15%.

A false positive rate this high introduces a significant amount of benign traffic into the dataset, which could skew observed distributions and patterns. In the process of our work, such cases were found as well and indicated high usage for a specific port, that turned out to be false. The authors of the original paper also validated their classifier against the network telescope utilized later in this work (see Chapter 4).

Comparing the networks the authors marked as dark/grey with the validated blocks of the network telescope shows, that their largest vantage point incorrectly identifies almost the entire actual dark space. When combined with data from other vantage points, they then incorrectly mark the active /24s as dark as well. While the proposed method is promising, the current implementation does not yet produce accurate enough data to substitute for a real network telescope.

Another recently proposed idea is using the vast IPv4 address pool owned by cloud providers [38]. Deploying compute instances at these cloud providers and capturing the traffic sent to their attached IP addresses allows for a globally distributed telescope with low costs. It also solves the issue of scanners learning and avoiding telescope addresses since they might also house normal cloud services. In fact, they show that they receive 450 times the traffic that they would expect for the amount of addresses they listen to.

However this also showcases a big problem with this system. Since cloud provider addresses are not dark and mostly used by legitimate services, changes are high that they receive stray benign traffic. In fact, the same authors have shown in their previous work, that AWS compute instances receive traffic meant for services deployed at those addresses previously [39]. Based on results of that work, they filter the traffic analyzed for the DScope telescope. As a result, 54k source IPs or over 25% of source IP addresses are removed. After the filtering they only observe 125x instead of 450x the traffic they would expect and their variance in seen source IP addresses drops to 14.5, almost equaling the metric of the reference network telescope and directly invalidating one of their findings (F2).

As indicated by the sharp decrease in observed source IP addresses after sanitation in their paper, one cannot discard the impact of IP reuse traffic on overall telescope data. The filter methods employed are also not very sophisticated and might remove AWS deployed scanners but let traffic that contacts more than one AWS IP address through. Both scenarios are likely to happen, especially given longer observation periods.

This highlights the difficulty of using active IP addresses. Even with the filtering they still see more traffic than the reference network telescope. That is great for collecting additional data but could also mean that the IBR collected by cloud deployed network telescopes is not fully representative for non-cloud hosts.

2.4 Packet Loss in Transit

Packet loss can occur at many points along the path of a packet for many different reasons. Figure 2.5 shows the origins considered in this work. Sources that intentionally avoid the telescope are not part of the scope of this work. Due to the very heterogeneous device landscape within the Internet and traversal of many ASes, both intentional and unintentional loss are possible. Intentional packet drop might for example be caused by a firewall, while unintentional loss is could be caused by an overloaded router. When trying to reach conclusions from IBR captured at a network telescope, the possible impact of such loss should be considered. If not corrected, distributions and number of occurrences might be skewed.

2.4.1 BGP Hijacks

BGP hijacks incorrectly route traffic to a different AS. While the packets are not necessarily lost in the sense of being dropped on path, they are lost for the original destination. A sender will not receive an answer, since the intended receiver is not reached by the request packets. An exception is if an

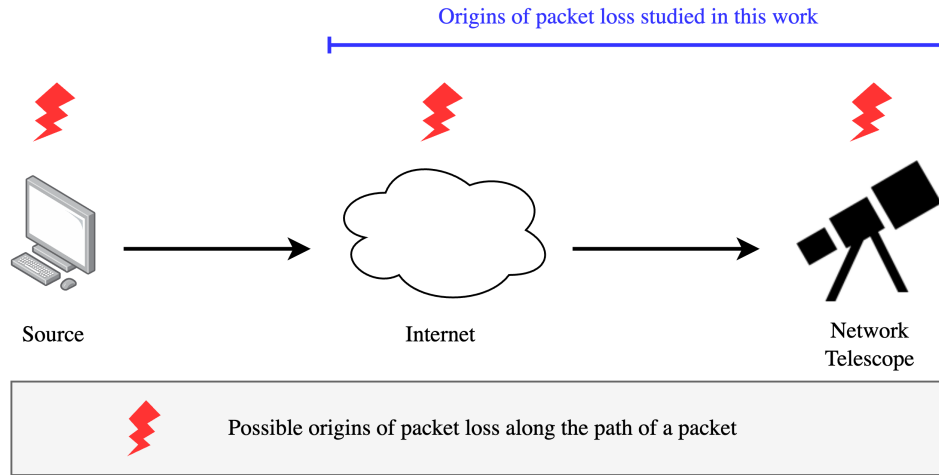


Figure 2.5 Origins of packet loss along the packets path. In this work, losses after the source are considered.

attacker places it self in the AS PATH but forwards the packet to the correct destination later. A reason for forwarding the packet afterwards might be to eavesdrop on the communication between two networks without interrupting it. BGP hijacks happen quite often [12, 23, 44], however their global reach is usually limited, as the routes announced by the attacker get rejected by some peers, which stops propagation.

There is a multitude of reasons as for why BGP hijacks happen. Research has been done to identify the different reasons and discern malicious from accidental actors [8, 43]. While misconfigurations with typos and wrongly applied path changes account for many events, not all can be explained with those. In some cases correlation with other observed phenomena, such as increased spamming, allows for more concrete attribution [49].

Attempts to prevent unauthorized announcements and their propagation have been made mainly in the form of “Resource Public Key Infrastructure” (RPKI) [26]. With this system, the right to announce a certain prefix by an AS is attested via asymmetric cryptography by the owner of the prefix. This attestation is verifiable by other parties. In concept, this allows a BGP router to verify whether a route received in an announcement is legal or should be discarded [30]. In order to protect the ecosystem entirely from route leaks and hijacks however, RPKI would have to be implemented by every single AS. The adoption rate is growing, with many big networks participating, but a global roll out has not been achieved yet. Due to the participation of large ASes and transit providers, the impact and propagation of rogue announcements has declined in recent years [28]. A motivated attacker is however still able to overcome RPKI in some cases [20].

2.4.2 Middleboxes

Middleboxes are intermediate nodes along the path of a packet. They serve many different purposes, spanning from network address translation and performance optimizations to security concerns. Sometimes, intentional dropping of packets is an essential part of the role they fulfill, but other times they may interfere with traffic in unexpected and unwanted ways. When trying to interpret IBR, we need to be sure that no specific type of packet is lost in transit, since this would influence our observations.

Security focused appliances such as firewalls and intrusion detection systems may drop packets they deem malicious or against set rules. Additionally, these devices might strip parts of a packet or outright discard it, if they include unknown values. Examples for these may be TCP options set by relatively new TCP extensions of which the device is not aware of. There are also variants of these with payload inspection that apply actions like discarding or rewriting packets. As a result of

these systems fulfilling their role, they influence the amount of circulating malicious packets. If this happens on a path to a network telescope, then false conclusions about the overall amount of bad traffic or the origin AS may be reached.

Another class of middleboxes are those providing “Network Address Translation” (NAT). Such a system takes packets from a network A, replaces the source IP addresses with its own assigned address in network B and sends that packet via network B. It also keeps track of the ports that it sends these packets from. That allows the NAT box to forward response packets back to the origin device. This is often done to give multiple devices in a private network access to the Internet, without assigning each individual device a publicly routeable IP address. NAT is the main solution to address sparsity, by allowing many devices to use the same IP address. Since system employing NAT have to interact and modify packets, they might run into the same issue of having to cope with unknown protocols or extensions. Another side effect of NAT is that traffic that might seem to originate from the same device is in fact coming from multiple systems behind the same NAT box.

Depending on the setup and attributes of certain networks, performance optimizing devices may be deployed. Applications like “Performance enhancing Proxies” (PEPs) and “Traffic normalizers” interact with packets and flows on the transport layer. PEPs can influence packet flows by sending early ACKs to speed up a connection or perform retransmissions in the name of the actual source. A traffic normalizer will capture packets with out-of-order sequence numbers and forward them in the correct order. Additionally he might change TTLs within a flow to be the same for every packet. While these devices do not drop packets, they change the content within a packet or send out additional ones.

2.4.3 General Packet Loss

Besides malicious actors and devices intentionally dropping packets, traffic can be lost due to buffer restrictions or hardware failures. General packet loss within the Internet varies a lot between different source and target ASes. Both the connectivity and geographical distance between the two networks or intermediates affect loss rate. Wan et al. places the general average packet loss at 0.44–1.6% for their scanning activities from different vantage points [51]. They specifically note Australia as having higher packet loss due to the generally worse connectivity.

There is a number of possible causes for general packet loss. The first is link saturation, where a router along the path of the packet experiences high loads and as a result starts dropping packets. Depending on the configuration, they might employ mitigation techniques like for example “Random Early Discard”, where the device will start dropping packets even before an interface is fully saturated. Of course these situations are not desirable and in cases of transit providers there is a valid business case for making sure this does not happen. However, given lead up times for upgrades and unexpected traffic spikes, they still happen. This also applies for provider backbones [29].

In some cases misconfigurations, like BGP route leaks, can also cause links that are normally sufficiently equipped to be overloaded. An example is a route leak from Telekom Malaysia on June 12th 2015, where its upstream provider Level3 accepted incorrectly announced routes. This caused traffic spikes at some Level3 PoPs, as many peers now sent their packets via these rogue routes. These traffic spikes in turn led to link saturation at these PoPs impacting customers of Level3, even if their traffic was not rerouted by a rogue route [24].

When changes in the routing layer happen, they have to be propagated and accepted by all involved parties and routers. While that happens, packets may be sent along routes which are no longer available. Traffic like that will potentially not reach the destination. The results are longer loss events [29], where a target host might even seem offline [51]. This applies both to intra- and inter-domain routing changes, as packets at an intermediate AS often take a few hops in that network until the next AS is reached. Older research suggests that most bottlenecks like these are located in Tier-One networks [1], however this might be different today.

To underline the various places and reasons for general packet loss in the Internet, research shows that even the weather can have significant influence on the number of downtime events [37].

3 Measurement Design

Figuring out what traffic **does not** reach the telescope, requires knowledge about what traffic **should** reach it. By searching for the known traffic in the recorded network telescope data, we can identify missing packets and perform further analysis on probable causes.

Sources for this known traffic can either be active or passive. For active sources, we control the packets sent to some extent, while passive sources utilize observed real world traffic originating from outside the measurement setup.

3.1 Traffic Collection at Network Borders

Passive traffic sources are the preferable way for testing the telescope as no new packets are inserted into the dataset of the NT. Measurements therefore do not skew the original dataset. While there are ramifications that could be deployed by a NT for active measurements, like filtering, this involves additional setup overhead. Another advantage of passive sources is that they cover real world cases. The traffic captured by a passive source contains use cases and patterns actually appearing in the wild.

There are four major properties of vantage points (VPs) relevant for this work:

1. Traffic volume
2. Resolution or Granularity
3. Composition of traffic
4. Location on packet paths

3.1.1 Amount of Traffic Logged

Traffic volume describes the amount of traffic processed and therefore observed by a VP. The main factors for volume are the market the VP operates in and the size of its customer base.

Resolution tells us the granularity of the logs collected. It is often a trade-off between the amount of data stored and the storage cost for that data. An obvious way to log traffic is to just store every packet observed, however this will incur huge storage costs. For that reason, parts of a packet are often discarded, for example the payload gets truncated.

Flowdata or FlowTuples take this idea one step further by combining a set of packets into a single “flow”. Since most communication over the Internet is made up of requests and responses, entire sessions can be efficiently stored in a single flow. A lot of data within a typical connection is redundant and only has to be stored once. The exact fields kept in flow data depend on the implementation but

important and common information like source and destination IP address and port, as well as the used protocol are generally included. For data that changes between packets in a single connection, like packet sizes or TTLs, aggregates like averages or the most frequent value are sometimes provided while individual values are lost. For the goal of this work, flow data is not optimal, as we will try to match packets from our sources with packets collected by the telescope. Missing parts of packets make it harder to perform this matching accurately.

Another way to optimize storage cost is to not store all data but take samples. A sample rate of n then generally means that every n th packet or flow gets stored. With a higher sample rate the relative traffic volume that actually got stored by the VP falls.

In a perfect world VPs would both have high volume and store non-sampled packet-granularity logs. In the real world however, this is often not financially feasible.

3.1.2 Potential Bias of Vantage Points

The composition of traffic observed by a VP differs based on what networks send packets through it. If a VP mainly handles traffic from eyeball networks like local ISPs, one could expect more incoming traffic and certain protocols like HTTP(S) playing a larger role. On the other hand, if traffic processed comes from data centers, more outgoing traffic is probable. Certain networks containing specialized equipment might use rare ports for example. Depending on the geolocation of a provider there might also be temporal or seasonal patterns that differ from other regions. Events held in specific parts of the world might result in a larger share of some protocols, which makes the composition not representative of the entire Internet. The best case for a source like this would be a global transit network, that generally should observe all types of traffic with a low vulnerability for local spikes and events changing the distribution significantly.

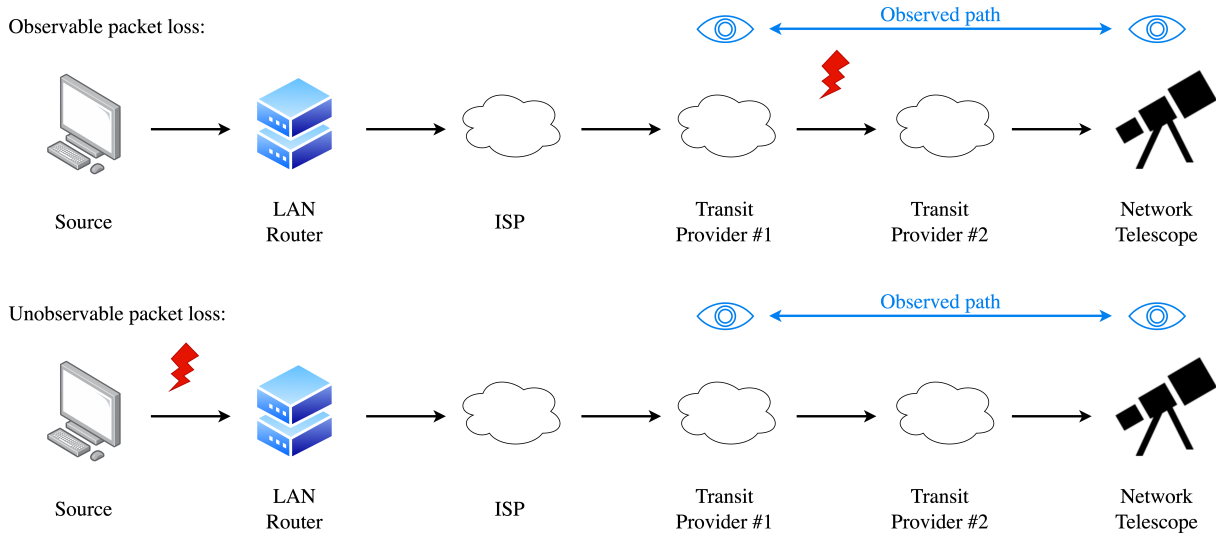


Figure 3.1 Example path of a packet with different possible locations for packet loss (lightning).

Transit Provider #1 is the VP, so the path between it and the NT is observed (blue arrow).

For another criteria such a big provider is however the least optimal VP. When thinking about where packet loss likely occurs, the path between a Tier-One transit provider VP and a NT does not seem likely. With most network telescopes being very well connected in order to capture as much traffic as possible, the path between a transit VP and the NT only includes other Tier-One providers. That connection is still subject to overload and interface saturation as will be shown in chapter 6, but these would also be covered by any VP earlier in the path of a packet. Middleboxes that might cause loss have mostly been traversed when packets reach the transit provider, so any lost traffic will not show up in its logs and can therefore not be detected. As seen in Figure 3.1, a VP can only observe packet loss (indicated by the red lightning) when it happens between its position and the telescope,

indicated by the blue arrow. This is shown in the top path. The packet loss shown in the bottom case is not observed, since the packet will never reach the first detection position. This highlights the difficulty and complexity in finding data sources suitable for this work.

3.2 Active Probing

Besides collecting traffic at network borders, active measurements can also be utilized. These allow for more control over the traffic sent. Different types of measurements allow for different insights. When working with the telescope, the integrity of its collected data should always be kept in mind.

3.2.1 Type of Measurements

Like most scanners observed by network telescopes, one can perform simple port scans using either TCP or UDP. Here, one or more packets of the targeted protocol and port are sent. In case of TCP scans, this is often just an empty SYN packet, looking for a handshake response. For UDP scans, the packet might contain an application related payload, so that the service potentially listening on the target port will respond.

If there is no exact timing data available from the source, its hard to match observed port scan packets in the NT with expected ones. This is due to the fact that a single measurement of this kind itself has no detectable pattern.

Measurements sending multiple packets are better for analysis and are therefore preferable. This is due to the fact that a running measurement can be indicated by the first packet reaching the telescope. No exact timing data from the source is required. To check for packet loss, one can then check the received number of packets and compare them against the expected value. While this will catch most small packet loss, its not possible to observe an entire measurement getting lost. For random packet loss, this gets more unlikely the higher the number of packets in a measurement. Loss of a full measurement might indicate a middlebox on the path. In order to detect these events, coarse time data (*i.e.*, expected number of measurements per day) is required.

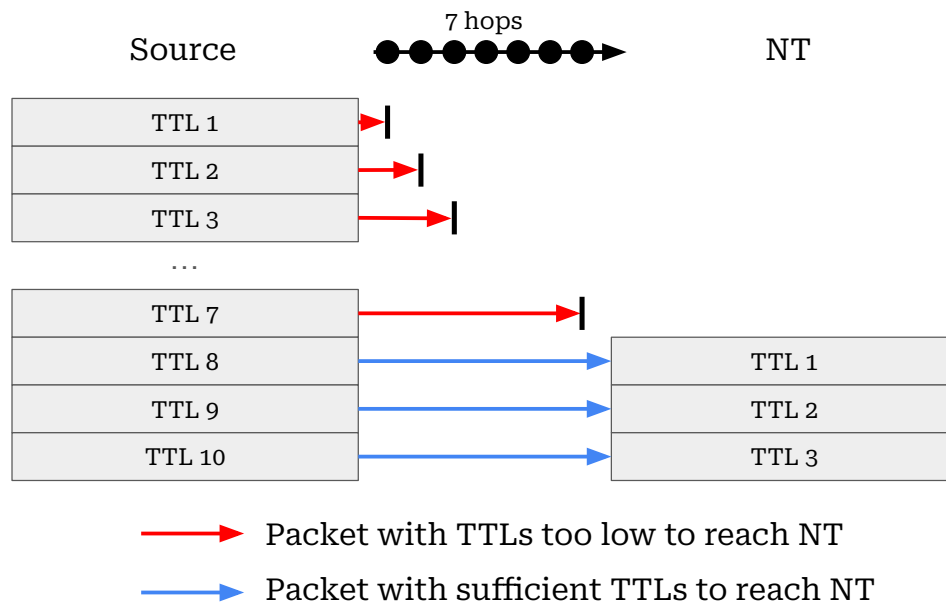


Figure 3.2 Visualization of a traceroute arriving at the NT

“Pings” are a kind of multi packet measurement. A default ping performed by the likewise named

Linux tool sends multiple ICMP packets of the type ECHO REQUEST. Even with the telescope not responding to these requests, all are sent. This might differ for specific sources and implementations, with RIPE Atlas using three packets as a default value. Pings are however limited to the ICMP protocol and can therefore not cover all kinds of traffic observed in the wild.

A measurement providing even more information is a “traceroute”. By increasing the TTL in the IP header by one for every successive packet, the sender tries to probe the path towards the target. Since a host should respond to a packet with a TTL of 1 with an ICMP packet of the type “TTL EXCEEDED”, the sender can uncover the IP address of an intermediary hop. When a traditional network telescope gets targeted by a traceroute, it will not respond. This generally results in more packets with higher TTLs being sent to the same telescope address.

Figure 3.2 shows the pattern of the packets received by a NT when targeted by a traceroute. In the case of this example, a traceroute with a total hop count of ten was performed. The path from the source to the NT is seven hops long, which results in three packets being captured. By using the maximum TTL received, the total number of packets expected can be inferred. Additionally, if packet loss occurs, the position of the lost packet is known. All without any timing information from the source.

The TTL value can not be trusted fully however. Path changes that are bound to occur even within a measurement can result in a different path length. This in turn effects the maximum TTL. If a path shortens within a measurement for example, using the maximum TTL and counting the packets would indicate loss even though all packets were received. In the same way, a longer path can mask loss. Reasons for path changes are load balancing and traffic shaping, which can happen at any point on a path. Routing changes can also result in different path lengths.

In order to stabilize the path of a traceroute, one can use “paris-style” traceroutes [4]. This technique keeps headers used for load-balancing constant between packets, resulting in packets being sent via the same path. While this can eliminate most inconsistencies with traceroutes, it is not a complete solution.

3.2.2 Third Party Measurement Projects

Cooperating with external scanner operators has several advantages. The data of the network telescope will not be influenced by the experiment. These scanners would perform their scans regardless and are “true” IBR, which rightly should be contained in the data of the NT. By using these projects instead of additional active measurements, we can reduce our measurement footprint. If the scanning sources, targets and patterns are known precisely, one can determine the exact number of packets expected in the NT.

A major difficulty is of course to find scanning projects willing to cooperate. The methodology of this work requires quite intricate information of scanning setups and schedules. This includes source IP addresses and the type of measurement. To understand potential causes for packet loss in the analysis phase, additional communication and cooperation may also be required.

Another issue with using specific traffic sources for analysis is that the traffic may not be representative of general IBR collected by the NT. While scanners make up a significant chunk of received packets, there are still different types and parameters associated with each scanner. Most operations do not perform every scanning activity but rather focus on specific business cases. As a result, they may for example use certain ports or protocols. This opens up blind spots in the analysis if one relies solely on this type of traffic source. Overall conclusions can hardly be drawn from a single project.

This gets amplified by the low number of sources for a single scanning project. Even though an operator might utilize multiple scan hosts and topological locations, often these are fixed addresses with fixed paths via the same transit providers towards the NT. From the perspective of the operator, this is an advantage, since external factors can be limited. For this analysis that is also true when trying to establish a benchmark or ground truth. It is however not a diverse enough data base to reach overall conclusions.

3.2.3 Self Initiated Measurements

Instead of using other scanning projects as traffic sources, one can also perform their own measurements. By owning the setup, methods of measurement are not limited to the ones preferred by a scanner operator. Still, the danger of blind spots remains, as not every protocol and parameter combination can be tested due to the sheer amount combinations. A unique capability with active measurements is that cases which showed interesting behavior in other data sources can be supplemented with additional measurements.

With active measurements, additional work on the side of the network telescope may be required. Since the resulting traffic is not organic IBR, it itself skews the data collected. Active scanning campaigns therefore need to be tracked by the telescope, so that mitigations like filtering or announcement to other researchers can be put in place. The effort is increased for more complex scanning operations spanning more source and destination addresses.

A property preferable for the planned measurements is diversity of sources. To increase the likelihood of encountering non-random packet loss, as many paths as possible should be covered. This is not limited to just different AS paths towards the NT but includes devices on path. Therefore, systems close to end users are optimal, since the probability for a middlebox on path is highest here. Many botnets utilize poorly secured IoT devices or routers. Their traffic, which is interesting for researchers, therefore also originates in these networks.

Acquiring scanning nodes like that is not trivial. Internet measurement projects like CAIDA Ark or RIPE Atlas are likely the best candidates. With probes deployed in external networks without any control however, there might be external factors that can influence packet loss beyond the scope of this work. Replacement of networking equipment or flaky connection types like wireless systems are just two examples of interruptions that can occur. Since probes are administered by external personnel, debugging these problems can also be challenging.

Overall active measurements are still a great way to cover many paths with different measurements. The uncertainty and complexity involved however requires a higher level of scrutiny when analyzing the resulting data. In combination with data collected at a large VP, these sources can be used most efficiently.

3.2.4 Summary

To understand overall traffic patterns towards the telescope, high volume vantage points like Tier-One transit providers or large IXPs are indispensable. They handle large amounts of traffic, so the data is very representative. Complete packet collection would be preferable, although it is unlikely at this scale. These vantage points form the base for packet to packet matching.

To allow for verification and the creation of ground truth, external scanner projects can be used. These offer a control group for the reachability of telescope addresses in a certain time frame. A constant inflow of packets to every telescope address implemented by an active measurement campaign can only be done with great care, as to not influence the NT data too much. External scanners provide a simpler way in this case, since they are regular IBR.

Lastly, self initiated active measurement campaigns should be used to investigate specific types of sources and to get as close to potential middleboxes as possible. While they cannot cover the entire spectrum of different protocols, ports and patterns, probes like those provided by RIPE Atlas allow for very fine grained analysis.

4 Measurement Setup

4.1 Traffic Collection

Traffic collection was performed by the CAIDA Network Telescope, which provides different datasets derived from captured IBR.

4.1.1 CAIDA Network Telescope

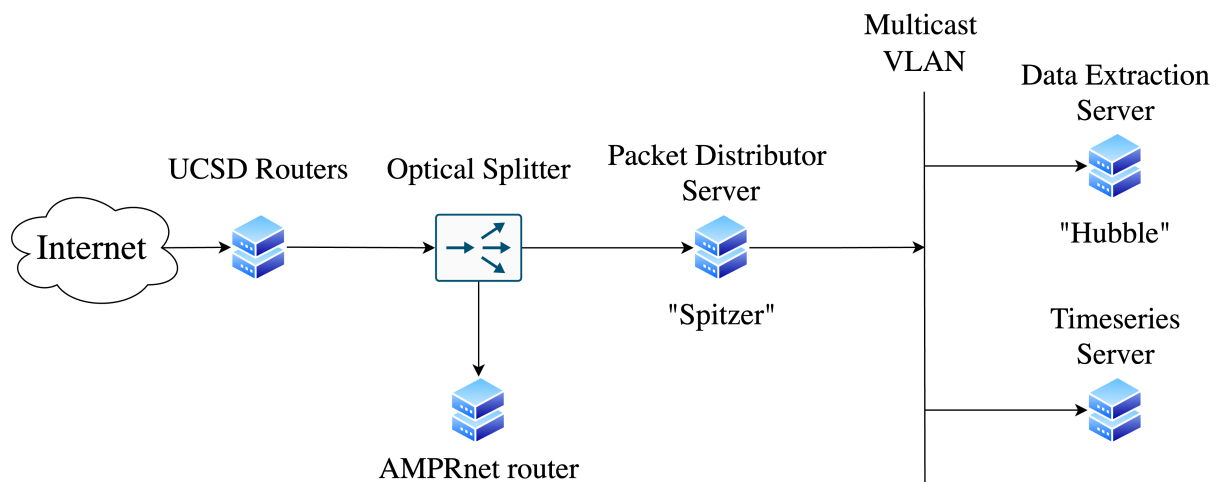


Figure 4.1 CAIDA Network telescope capturing architecture [7]

The Center for Applied Internet Data Analysis (CAIDA) stewards the network telescope located at the University of California San Diego (UCSD) Supercomputer Center. It captures traffic destined to a /9 and a /10 prefix. The address space used is announced since 2003 but has remained mostly unused. The UCSD receives its uplink from Level3 (AS3356) and Hurricane Electric (AS6939) via AS2152, the “Corporation for Education Network Initiatives in California” (CENIC). The prefix is well visible, with RIPE Stat recording a 100% visibility in participating route collectors.

The prefix used by the telescope, once an entire /8, is owned by AMPRNet. They allow operation of the NT on unused IP addresses. Some more specific prefixes are leased out to individuals, making them unavailable to the network telescope. A filterlist for these used prefixes is maintained by CAIDA. This list gets created by querying two API endpoints provided by the organization managing AMPRnet. One of the endpoints returns all allocations provided to members for use in an internal mesh network. The other one provides a list of BGP announced prefixes, as lease holders sometimes announce their more specific prefixes in BGP with their own AS.

Figure 4.1 shows the architecture of the network telescope. Since the UCSD announces the prefixes of AMPRnet, its routers receive all the traffic sent to addresses within the prefix. Packets then get forwarded to both an AMPRnet router and to the “Packet Distributor Server” (Spitzer) via an optical splitter. While the AMPRnet router forwards packets to the internal mesh network, Spitzer applies the filterlist to remove that active traffic.

All traffic that is not filtered out gets packed into jumbo frames and broadcasted into a local VLAN. Any system that further wants to utilize the traffic listens to a multicast address in that VLAN. While researchers can get direct access to this VLAN, the two most important systems attached to it are the “Data Extraction Server” (Hubble) and the timeseries server.

Hubble collects all packets from the VLAN and derives the PCAP, Flowtuple and RSDoS datasets provided by CAIDA from those. The timeseries server is responsible for computing the aggregated time series datasets.

4.1.2 Types of Data

The telescope currently handles around one million packets per second with peaks of over three million packets per second. With an average amount of ~ 400 Megabits per second, raw traffic stored per hour amounts to ~ 150 Gigabytes when compressed. Storing these large amounts of data is no trivial task and becomes expensive for longer periods of time.

Raw captured packets in the PCAP format are stored for 35 days. These are provided via a S3 bucket as large compressed files, each containing one hour of traffic. This gives researchers the highest amount of resolution possible. However, searching through one of these files, which can grow to over 200 Gigabytes, takes over an hour. This type of data is the main source used in this work, as we try to perform packet to packet matching.

Parsed from the stored PCAPs are flow tuples in the Apache Avro format. Since these flows are much smaller, they are cheaper to store. Flowtuples are available from August 2008 onwards. Since these are created from the stored PCAPs, they were not used in this work, as they provide no additional information.

CAIDA also provides a Grafana dashboard containing time series aggregated from telescope traffic in near real time. This data gets created independently, so its availability is not based on stored PCAPs. These time series provide coarse telescope wide information like current packets per second or bits per second. They however also track more fine grained information, like the distribution of protocols and ports observed. Additionally, many of the metrics are provided on a per country or AS basis. We used this data source to validate observations made in the PCAPs as well as identify phases in which the telescope was experiencing exceptionally high loads.

4.2 Traffic Sources

We utilize three different types of sources to collect traffic moving towards the NT.

4.2.1 Traffic Collection at Network Borders

For collecting general traffic directed to the address range of the NT, we use traffic and flows sampled at two network vantage points.

The first one is a local german IXP. They provide truncated packets sampled from the traffic of their members. While they do not see a lot of traffic with destinations in the NT, the packet level sampling allows for a more detailed inspection. Having many details like TCP options and parts of the payload makes this source highly valuable.

The second source collects sampled flow data. It is a globally operating Tier-One transit provider handling upwards of 50 Tbit/s of traffic. The flows are collected on the edge of the AS, when packets from peers enter it. Due to the enormous amount of data handled, each flow is usually made up

of only a single packet. This is because the store for tracking ongoing flows gets flushed by new flows before a second packet can be observed. This makes the source less optimal for packet level matching but non the less it is a valuable insight due to the amount collected and the general exposure. Although they operate globally, most traffic sampled originates in Europe, as most of the customer base is located there. The provider reaches the CAIDA network telescope via its peer Level3 (AS3356) and the AS Path is stable. A packet observed at the VP will therefore only travel through another Tier-One network before it is handed off to CENIC and the NT. The path is unlikely to contain any middleboxes or high loss links.

Table 4.1 Overview of Vantage Point sources

Source	Abbreviation	Type of data	Traffic collected	Timeframe
IXP	VP1	sampled packets	3 393 packets	01.08.23 - 30.09.23
Transit provider	VP2	sampled flows	310 227 flows	16.10.23 - 01.11.23

4.2.2 Known External Scanner Activity

For this work we utilize two research scanning operations, which provided us with details about their setups. From this, we can derive expected packets at the NT.

One of these sources continuously probes the entire IPv4 address space, checking popular ports for each host. The exact period for each full scan is not known, but within the time span of a week, every address of the NT had over 10 packets recorded. This source is a great ground truth provider for general NT availability. Some of the traffic of this source is routed via VP1, so it can be observed on multiple points along the way. We will call this source SC1. Its scans are concentrated, with down times between individual scans.

The second scanning activity (SC2) does not probe every single host, but rather checks a random host within every routable subnet of a certain size. The selected host gets probed by a set of twelve different sources, which each are provided an upstream link by a distinct AS. These connectivity providers are mostly Tier-Ones, but also include some smaller regional providers. Not all sources send their probe at the same time, rather they are distributed over a full day. When analyzing the packets of SC2 in the following chapters, a single traceroute from a single location will be called a “measurement”. Due to the implementation, the distribution of measurements results in a relatively constant number of measurements per minute.

Besides the many transit ASes involved and the resulting diversity in paths taken by individual packets, this second scanning activity has another advantage. Each measurement from a particular source is a paris-style TCP traceroute with a fixed maximum TTL of 25.

Table 4.2 Overview of Scanner sources

Source	Abbreviation	Type of data	Traffic collected	Timeframe
Port scanner	SC1	TCP SYN scans	631 829 522 packets	01.09.23 - 30.09.23
Multihomed scanner	SC2	TCP traceroutes	43 997 831 packets	12.11.23 - 26.12.23

4.2.3 RIPE Atlas Probes

The RIPE Atlas Project is a distributed network of probes that allows users to perform a multitude of different types of requests. It consists of over 12 000 probes, operated by volunteers, located in over 179 countries. A measurement can include pings, traceroutes and application layer protocol operations.

For this project, we try to leverage the global span of the network, with more than 3 000 IPv4 ASes covered. By using as many different paths towards the NT as possible, we broaden our view

on potential packet loss. Atlas probes however come with some caveats. Since they are operated by volunteers often not associated with RIPE, network quality and probe uptime can vary. Possible connection types include wireless networks like cellular or satellite, which are inherently more prone to packet loss. Depending on the setup of the network the probe is situated in, some measurements might also not work correctly. For example, some networks drop certain ICMP packet types, which breaks traceroute measurements. Additionally implementation details of the RIPE Atlas platform complicate the calculation of expected packets in the NT telescope.

For the tests run in this work, 250 probes were selected at random by the RIPE Atlas platform. The probe set covers 50 countries and 175 different host ASes. Currently the platform does not support creating distributed sets of probes. Therefore the selected part of the entire probe population reflects the uneven share of host countries, with origins like Germany and the US being present more often. While a more diverse selection process could be implemented independently, the selected set was deemed distributed enough, especially in the face of the many different ASes. Figure 4.2 shows the AS distribution, while Figure 4.3 shows the country distribution with different colors inside a single country indicating multiple ASes.

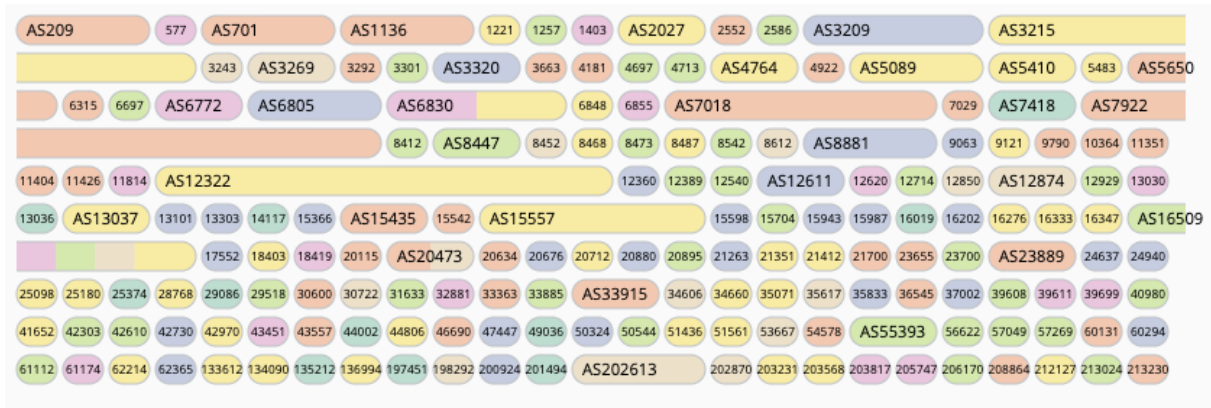


Figure 4.2 AS distribution of probe set as reported by RIPE Atlas. The size of each AS represents the number of probes participating from that network.

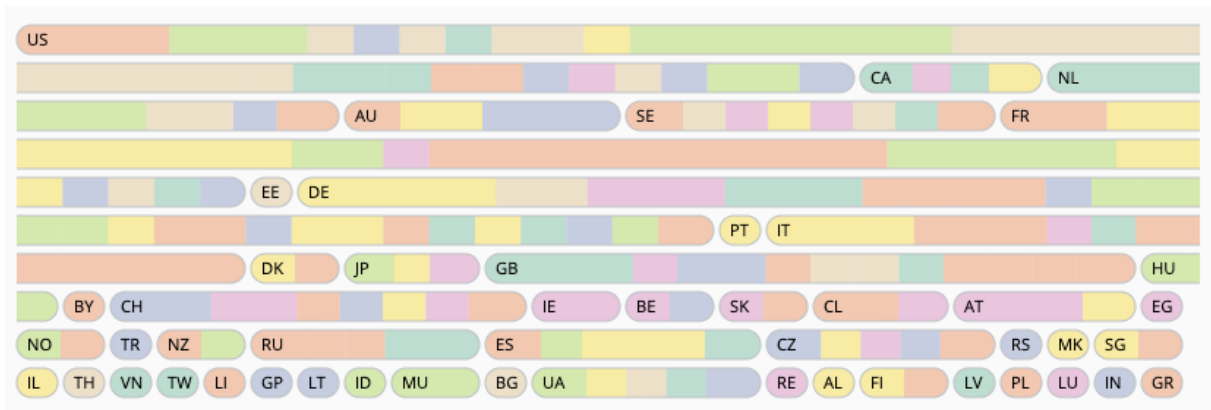


Figure 4.3 Country distribution of probe set as reported by RIPE Atlas. The size of each country represents the number of probes participating from that nation.

The operator of a probe can attach tags to his probe, which may include the connection type, the type of network the probe is deployed on or whether the IP address comes from a NAT device. However, these user provided tags are not mandatory or audited, so they are in no way a guarantee. Tags prefixed with “system-” are set by the RIPE Atlas platform and can be trusted. System tags only cover some attributes. For the set of probes used in this work, Table 4.3 shows the required and

excluded tags.

Table 4.3 Tags used in the selection of the RIPE Atlas probes

Tag name(s)	Status	Set by	Explanation
ipv4	required	Operator	The probe has an IPv4 connected interface.
system-ipv4-stable-30d	required	System	The IPv4 connectivity was stable in the last 30 days.
satellite, starlink	excluded	Operator	Tags for probes having a satellite connection.
3g, lte, 4g, 5g, mobile	excluded	Operator	Probe is connected via a cellular connection type.
known-ipv4-issues	excluded	Operator	Probe sometimes has connectivity issues.
wi-fi	excluded	Operator	Probe is connected wirelessly to network.
ipv6-only, no-ipv4	excluded	Operator	Probe does not support IPv4.
system-ipv4-doesnt-work	excluded	System	Probe cannot complete any IPv4 measurement.
system-wifi	excluded	System	Probe connects via wifi.
system-flakey-power	excluded	System	Probe frequently shuts down.
system-flakey-connection	excluded	System	Probe frequently has connectivity issues.
system-bad-firmware-signature	excluded	System	Probe might have corrupted firmware.

From the set of probes we performed multiple measurements with different parameters. This was done to account for middleboxes that might only interfere with certain packet patterns. All probes performed their measurements three times within an hour, at randomized points so that no traffic peaks would be created at the telescope. By having three measurements, patterns might be differentiated from random events.

Every probe performed paris-type traceroutes to an address within the telescope. For ICMP and UDP, one traceroute measurement was run each. With TCP, RIPE Atlas allows the targeting of a specific destination port. Three distinct traceroute measurements were performed, covering ports within different classes. The IANA manages a list of ports and their associated applications. The list only covers ports up to 49151, since the port range 49152-65535 is reserved for dynamic ports. Middleboxes might choose to drop packets with specific ports according to a set policy. Port 80 was chosen as a widely used port from the IANA list, as it is the assigned port for HTTP. TCP Port 44901 is still within the range of the IANA list, but has no associated protocol and is not used for any popular application. The last measurement targeted port 53724 which lies within the dynamic port range. All other parameters were left with default values.

Besides traceroutes, each probe ran two ICMP ping towards a different address within the telescope. While one was a normal ICMP ping without a payload, the second one sent out packets with a size of 1650 bytes. That is above the standard MTU of 1500 and most likely results in fragmentation along the path. This was to test if fragmented packets might show a different behaviour.

During the experiment, one probe went offline, resulting in only 249 actually participating. 61 219 packets from these probes reached the telescope and were further analyzed.

4.3 Data Processing

4.3.1 Extraction

In order to filter out the packets of interest from the large amount of data collected by the network telescope, the easiest way is to search for source and destination addresses. Additionally, a rough estimate for the arrival time of a packet is required, since the telescope stores its PCAP dump in hourly

Table 4.4 Scan regime for RIPE Atlas probes

Measurement Type	Protocol	Destination port	Packet size in bytes
traceroute	ICMP	-	48
traceroute	UDP	(chosen at random by Atlas)	48
traceroute	TCP	80	48
traceroute	TCP	44901	48
traceroute	TCP	53724	48
ping	ICMP	-	48
ping	ICMP	-	1650

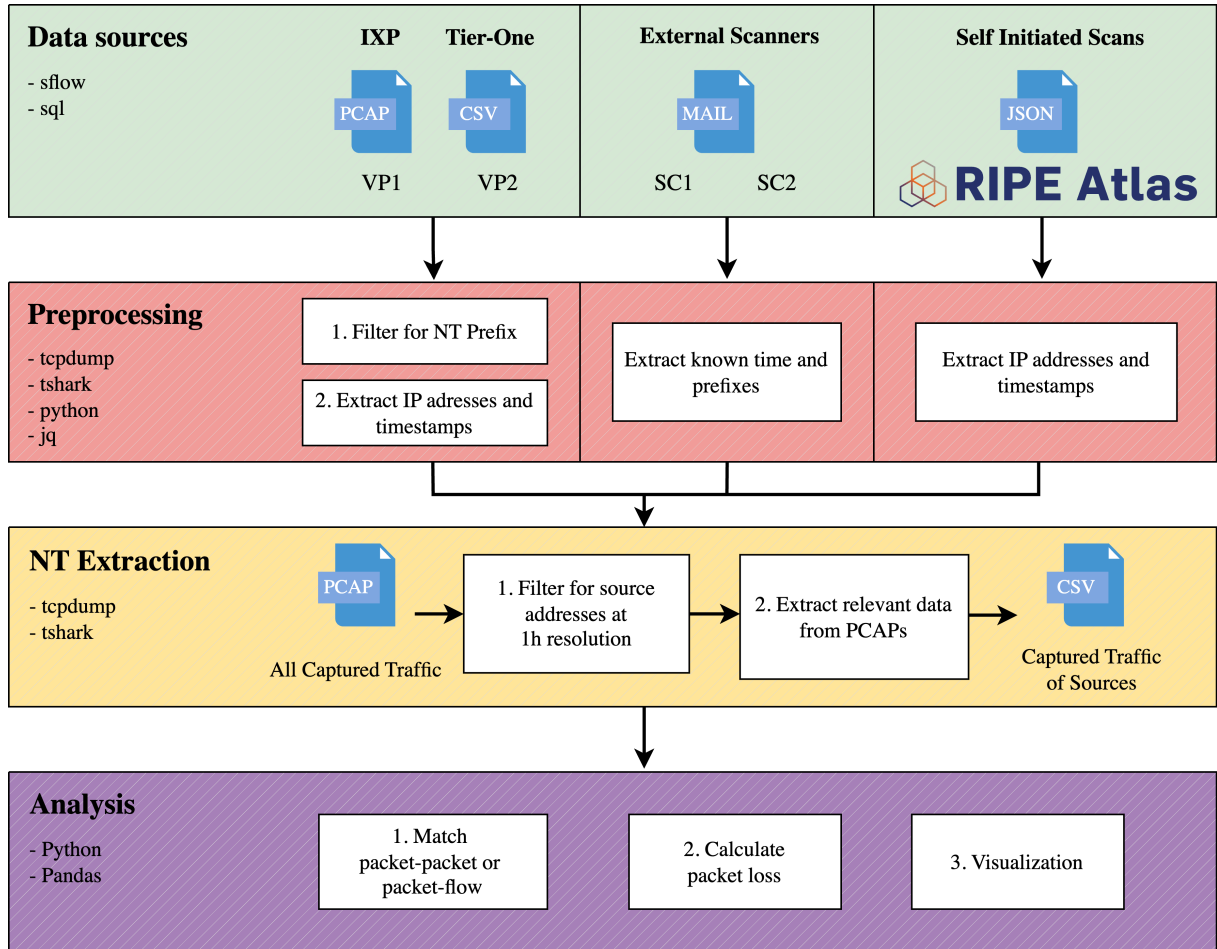


Figure 4.4 Data processing setup to search for lost packets.

rotated files. In order to find the packets we are looking for, we need to look in the right file. Figure 4.4 visualizes the entire data processing pipeline.

For the two vantage point sources VP1 and VP2, the collected data first has to be narrowed down to packets destined for the NTs /9 and /10 prefix. In case of VP1 with its stored packets in PCAP format, this is performed by the tcpdump tool. It allows for fast parsing of packets by utilizing eBPF. Using “tshark”, the command line version of Wireshark, relevant information from the PCAP files was parsed and stored as comma separated values. For VP2, which provides its flow data directly via CSVs, filtering was performed by a python script using the builtin “ipaddress” module. Afterwards the destination IP addresses inside the telescope prefixes are extracted along side the time that the packet was seen at. This allows for the selection of the corresponding PCAP file at the NT. If the time is five minutes or closer to the next hour, the next PCAP file will also be searched, as to not miss any

packets.

Collecting source IP addresses and timestamps is easier for the scanners SC1 and SC2. We were provided with the prefixes used for scanning by their operators. In case of SC2, meta data regarding upstream providers was given as well. As both scanning projects are continuous, no specific timestamps were provided. Rather, for performing analysis, time frames from one week to one month were chosen to observe the packets from these sources in the NT.

The results of a scan campaign using RIPE Atlas can be downloaded as a large single JSON file via the services API. From that file, all relevant information, such as traceroute hops and probe addresses can be extracted. For searching for the packets at the telescope, the probes IP addresses at the time of measuring, as well as the timestamps are extracted. The additional information will be used at later stages to figure out the number of expected packets and therefore the packet loss.

Resulting from this data processing stage is a large list of tuples containing source IP addresses and timestamps. Since the handling of a single PCAP dump file covering one hour also roughly takes one hour, source IP addresses are grouped by hour for efficiency. This way, each file only has to be parsed once by tcpdump, but potentially in multiple stages, as tcpdump has a limit on the number of rules it can apply. The smaller output PCAP file then gets fed into tshark which extracts relevant information from each packet and stores it as a CSV file. For feasibility reasons, all files are stored compressed.

4.3.2 Matching Packets from Vantage Points

Sources VP1 and VP2 provide data on packets in transit to the telescope. If they have reached it, the same packets should be stored in the corresponding PCAP dump file. To perform the matching, both the resulting CSV data from the vantage points as well as from the filtered telescope data are loaded into DataFrames, provided by the “pandas” python library. Loading the CSVs created from the NT output required additional work. The VM provided by CAIDA for research comes equipped with only 32GB of RAM, so the data has to be loaded in chunks and stored efficiently. Extensive usage of type conversion to smaller sizes and the category data type provided by Pandas made storing all data in RAM possible.

As a first step, vantage point data is further filtered with the help of the filterlist provided by CAIDA. Since the traffic to the addresses contained in the filterlist is not captured by the telescope, it would incorrectly be flagged as lost.

For VP1, which has sampled packet data, matching was done by merging on the source and destination IP addresses and ports as well as the IP protocol. This combination is also commonly referred to as a “5-Tuple”. Data collected at VP1 did not contain duplicate 5-Tuples with one exception. Additionally the time difference was taken into account, to further decrease false positives.

With VP2 data being flows, not much more information than the 5-Tuple and a timestamp is provided. After merging on the tuple, the time difference between the matched flow and packet is calculated. Only if the absolute time difference is smaller than five seconds, the match is accepted. The range was chosen to account for the travel time of a packet on one side and possible differences in local device times on the other.

A problem that arose during the processing of VP2s flows was that UDP traceroutes create a lot of matching misses, even though traffic has reached the NT. That is because for tracking individual hops, UDP traceroute increments the destination port with each packet they send out. If the flow sampled at an ingress router of VP2 is derived from a packet with too low of a TTL to reach the telescope, it will show up as a lost packet without adequate filtering.

This is true even if later packets of the same measurement do reach the NT and no packet that could have reached it, was lost. Since the destination ports of those packets sent for later hops are different than the packet sampled at VP2, they cannot be matched.

For the dataset used in the following analysis, two of the top five sources with packet loss were UDP traceroute scanners. It required additional sanitation to remove these from the set used for analysis.

4.3.3 Detecting Packet Loss in Scanners

The scanner SC1 performs scans over the entire address block of the NT. With the scans searching for open ports and vulnerabilities, usually only a single packet is sent. Checking for vulnerabilities generally means some form of banner grabbing, which will not work with a traditional telescope. So no additional packets are generated there either. Since no exact timing data was provided, looking for packet loss is not possible. Downtimes between campaigns also make it impossible to infer the time a packet should have arrived.

For this reason, only the reached IP addresses were extracted. When observed over a longer period, every single reachable address should have at least one received packet from this source.

For scanner SC2, received packets were grouped into single measurements. Since every source sends one traceroute a day to each subnet, grouping by source and destination IP address for every day is sufficient. Afterwards the TTL from each packet in a measurement is collected. If the number of packets does not match the maximum TTL or the list of TTLs is missing an intermediate value, the measurement gets flagged as lossy.

Time-to-live values can not be fully trusted, but analysis showed that maximum TTLs were stable for each source and average loss was almost zero. This indicates that the paths were stable. With SC2 performing paris-style traceroutes, these observations are in line with expectations. It can be assumed that traffic loss detected is mostly correct.

4.3.4 Detecting Packet Loss in RIPE Atlas Scans

For the Ping Measurements, packet loss detection is trivial. Every Atlas probe sends three ECHO REQUEST packets per execution, which should all be contained in the NT data. In case of fragmented packets, this number at least doubles. Here total packet length can be calculated within a run to determine if every fragment has reached the NT. Additionally, at least one packet with the “More fragments” bit set in the IP header and one without the bit set are to be expected.

Analyzing the traceroutes performed by RIPE Atlas is more difficult than for SC2. This is due to the fact that Atlas scans do not always send a fixed number of packets and that this fact is not documented. A maximum limit can be set, but possibly a lower amount is sent. For each TTL value three packets are sent simultaneously. The probe will abort its traceroute if for five consecutive hops not a single ICMP response is received. If an abort happens this way, three packets with maximum TTL are sent, called “last effort” by the Atlas source code.

With the prevalence of ICMP blocking or rate limiting, this did result in prematurely stopped measurements. For probes hosted in networks completely dropping ICMP responses, traceroutes are aborted after five hops. As a result, only the last effort packets can be found in the telescope. With many routers having ICMP rate limiting enabled by default, including the two routers handling telescope traffic at UCSD, every measurement has a unique number of packets.

Depending on how many routers in a row before the NT did not respond with a TTL EXCEEDED packet, a lower number of packets will be received by the NT. If all five non responsive hops are after the last router at UCSD, one can expect 15 packets. If the last router hit its rate limit and does not send out a ICMP message as a reaction to any of the three packets, only twelve packets will be received.

For all traceroutes not reaching 32 hops in total, the last effort packets have to be added.

All of this makes the loss analysis for Atlas complex. Due to the fact that none of this behavior is documented, extensive source code analysis of the RIPE Atlas probes was required to adequately interpret received packets.

4.3.5 Data Analysis

Analysis and comparisons were performed with the Python packages “Pandas” and “Matplotlib”. The most important visualization done is creating a time series plotting the packet loss observed.

The resulting graphs can both show the baseline as well as when peaks occur. Peaks then were investigated further, analyzing distributions of source and destination addresses as well as protocols and parameters set. As a last step, external data sources like the CAIDA grafana data source and BGP updates observed by RIPE were checked for these time frames in order to narrow down potential reasons.

The data was further sanitized by removing time frames in which the NT experienced exceptional load or other operational issues. These were detected through corrupted PCAP files and correlating loss peaks with load peaks from the CAIDA grafana data source. Additionally, PCAP file sizes were monitored to alert in case these were too small. A file size below 100 GB could indicate partial collection failure, usually at the beginning or end of the hour of data stored in a single file. Due to the high possibility of false positives and their relatively large impact for overall loss figures, extensive care in validation was taken. Time frames with known issues were excluded with padding both before and after. With losses often being below one percent of total observed traffic, low amounts of falsely detected loss can already heavily skew results. Still, all further numbers presented have to be considered as not perfectly exact but rather as upper bounds.

5 Operational Challenges of Network Telescopes

Operating a large network telescope in IPv4 comes with some challenges. Address space becomes more contested and the amount of traffic captured is ever growing. The CAIDA NT is no exception to this.

5.1 Inaccuracies of the Filterlist

During the measurement phase, some inconsistencies between the CAIDA provided filterlist and real world observations became apparent. To properly detect packet loss, these differences had to be investigated.

5.1.1 Understanding the Used Address Space

With the IPv4 address sparsity, it is difficult to obtain large dark blocks that can be used for network telescopes. The CAIDA NT, one of the largest publicly known telescopes, is located in a block commonly known as the “Amateur Packet Radio Network” (AMPRnet). The class A address block was allocated for use by amateur radio operators in 1981. It predates the public Internet.

The goal of AMPRnet is to provide addresses for the geographically distributed amateur radio community, so that communication can be based on IP. This allows the utilization of protocols built atop of IP like TCP and UDP via radio. The AMPRnet address block is further partitioned into regional subnets, as well as ones for independent BGP assignments or for management. To partake in AMPRnet, users can register with a foundation called “Amateur Radio Digital Communications” (ARDC) and request an IP address assignment inside of their respective subnet of AMPRnet. Leases are given out for a limited time, with the possibility to renew them. Allocations are not continuous. The entire management process is operated by volunteers.

Originally, nodes within AMPRnet were connected only via radio. Today, nodes that have Internet connectivity can also reach nodes where no direct wireless connection exists. They do this by using IPinIP tunnels. This protocol allows to encapsulate an IP packet into another IP packet. Through gateway nodes, that accept traffic from the public internet and then forward packets to their connected nodes, the inner packet with source and destination IP addresses from AMPRnet can be delivered. To achieve this, every member has to register the gateway that serves their assigned 44/8 IP addresses with ARDC. The list of all gateways and the members they connect then gets published by ARDC. Through this list, every member can “discover” all other members of AMPRnet and the path towards them. This effectively creates an overlay network over the Internet through which AMPRnet members can communicate with each other.

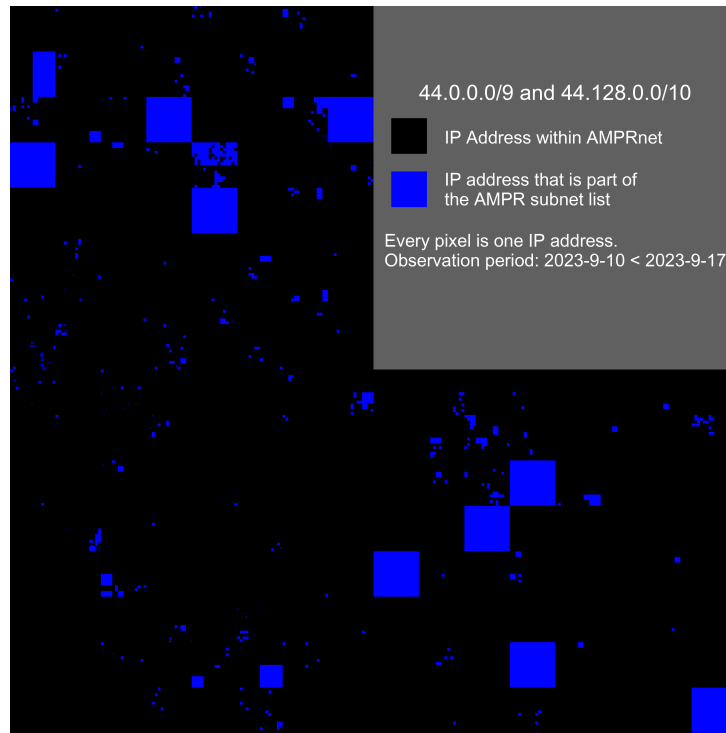


Figure 5.1 Hilbert curve of AMPRnet, IP ranges colored in blue are granted leases.

The reason for why the telescope can in cases receive legitimate AMPRnet traffic is that ARDC provides a general Internet-to-AMPRnet gateway. It receives all traffic coming from the public Internet directly to 44/9 and 44.128/10 addresses. If the destination address is a valid AMPRnet member and the packet is a well-formed IPinIP packet, then it gets forwarded to the member. It is this incoming traffic stream from the public Internet that the telescope is attached to via an optical splitter.

The important side effect of this overlay network is that utilization is invisible to the public Internet, with the exception of subnets that are announced independently in BGP by their lease holders.

Since 2001, CAIDA has partnered with AMPRnet to utilize the entire address block for a network telescope. For this, UCSD announces the entire prefix in BGP. CAIDA can then collect all traffic not destined to registered AMPRnet users via an optical splitter from the Internet-to-AMPRnet gateway. Since roughly 93.3% of the IP addresses within the block are not leased out, CAIDA can operate the NT on an address range worth over 300 million dollars without having to buy any IP addresses. Due to the fragmented nature of the leases granted by ARDC, the addresses belonging to the NT cannot be expressed in a single prefix.

In 2016 ARDC decided to sell a quarter of the address space to finance other activities of the foundation. Due to IPv4 address exhaustion, the sale raised over \$100 million. The upper /10 sold was not in heavy use and existing leases could be easily relocated. As a result, from the original /8 only the lower /9 and the remaining /10 are announced by UCSD in BGP. As a direct consequence, the NT address range also shrunk by 25%.

5.1.2 Filterlist Implementation

In order to differentiate between legitimate AMPRnet traffic and IBR in the packets collected after the optical splitter, the filtering machine (called “Spitzer”) uses a filterlist. This list is taken from the subnet list provided by ARDC to its members for establishing the mesh, which includes all currently granted leases. As it is not possible to differentiate between legitimate packets and IBR, all packets with a destination address in one of the contained leases are subsequently dropped by Spitzer. This list is regularly pulled, since new leases can be granted and old ones expire. Figure 5.1 shows the

distribution of leases granted and therefore areas blocked for the network telescope on September 17th 2023 as an example.

For this work, the filter list was used to more precisely filter for traffic towards the network telescope at the vantage points. Additionally, a second filtering run was performed on the extracted data at the network telescope as a precaution. The list was also frequently consulted when single high activity source addresses were found in telescope data.

5.1.3 Anomalies in Observations

After performing packet matching for VP1, initially packet loss in excess of 30% was observed. Suspiciously, the loss was contained to specific source and destination IP address pairs. These pairs had a loss rate of 100%, while packet loss for pairs which had at least one packet matched had a loss of 0%. Most source IP addresses were present both in successful and failed pairs. To validate the measurement setup, the total number of packets received by the destination IP addresses of those failed pairs was queried. The addresses in question had not a single packet stored in the NT for the entire observation period of one week. This is highly unlikely for an actual NT-IP, which usually receives at least around 250 packets per hour.

A quarter of observed traffic at VP1 was destined to a single IP address within the AMPRnet block. According to the filter list, there was no active lease, so packets should have been captured by the NT. When searching for the address in Shodan, a search engine for Internet connected devices, the IP address in question had an open port 443 which even presented a SSL certificate. A further search using the obtained certificate revealed that the node behind the IP address was operating as a TOR relay. It was registered with multiple sites tracking TOR exit nodes and relays.

Many destination IP addresses in the failed pairs were in the subnet 44.31/16. For this net, ARDC allows lease holders to announce their allocated prefix in BGP. In such a case, the telescope would not receive any traffic, as BGP routing prefers the most specific prefix. While BGP hijacks were a possibility, due to the nature of the 44.31/16 net, issues with the filter list were more likely. After contacting CAIDA, it was confirmed, that the filter list was no longer being updated. This issue was known and remediation was already planned. By switching the NTs filterlist to a new ARDC API, accurate and current data on leases should be used in the future. This change however only came into effect in mid November of 2023.

In addition to addresses in 44.31/16, there were also a few other addresses for which the NT had not stored a single packet. For these addresses no more specific BGP announcements existed and no entry in the AMPRnet lease list was found. To maybe identify patterns for these “dead” IP addresses and to at least create a trustworthy list of actual NT addresses, further examination was required.

5.1.4 Identification of Network Telescope Addresses

As stated in an earlier chapter, the known scanner source SC1 performs continuous scans of the entire IPv4 address space. It therefore also sends packets to every single address within the block used by the network telescope. While no scanning plan is known, given a large enough time span, every “Active Telescope Address”(ATA), should have at least one packet received by SC1. We will use these “marker packets” received by the network telescope over the span of one week to map the address space of the telescope. An IP address that does not receive marker packets will be called “Dead Telescope Address”(DTA), as these are of no use to the data collection of the telescope. More specifically, DTAs that are not in the filter list will be called “unexplained DTA”s (uDTAs).

Figure 5.2 shows the resulting distribution of unexplained DTAs. The smaller red areas are generally /24 in size. As visible from the figure, there are substantial areas of DTAs, which are not covered by the filterlist. The gravest offender is a /17 located in the upper left, as well as a collection of /24s in the center of the upper left quarter. That second area is the 44.31/16 subnet. Since it has already been established, that the filterlist is not up to date, we can add BGP data to the Hilbert Curve. There

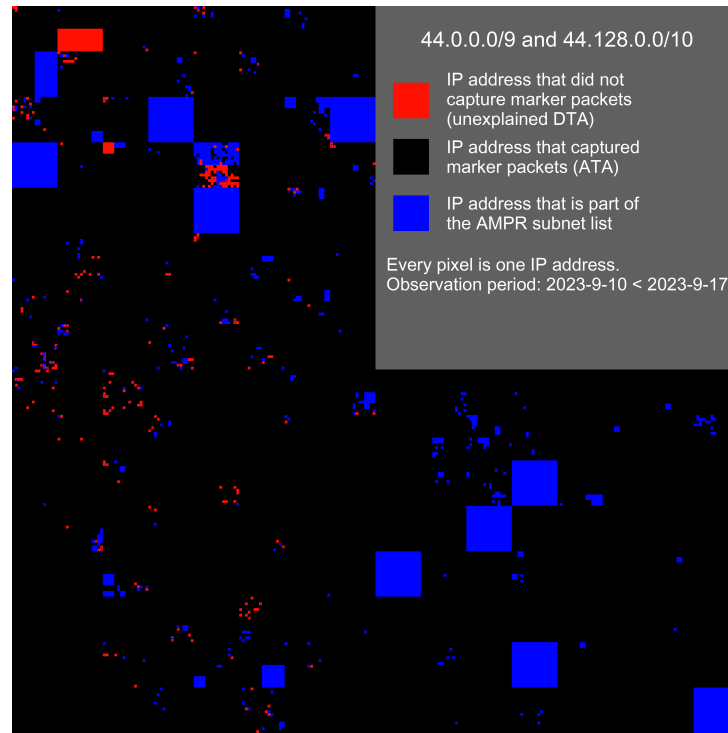


Figure 5.2 Hilbert curve of AMPRnet, where DTAs are colored red, ATAs are marked black and subnets from the AMPRnet list blue.

is no way at this point to differentiate between newer lease holders and genuine BGP hijacks, so any more specific announcement will be regarded as a granted lease for the moment.

Figure 5.3 shows the Hilbert curve enriched with BGP data. The 44.31/16 region shows no more uDTAs. This strengthens the theory that newer leases that are allowed to self-announce their prefixes, are responsible for the previously unexplained DTAs in that subnet. Still, there are more uDTAs scattered throughout the address space. An interesting observation is that within each uDTA square, there are at least a few IP addresses which appear on the subnet list. These red squares are at least the size of a /24. Since it is known that the capturing software used by the NT “parses” the subnet list and filters based on it, one potential explanation is the implementation of that filtering. 42.8% of entries in the AMPRnet subnet list at this time were prefixes more specific than /24, but the filtering seemed to imply that it only contains /24 or less specific prefixes. To test the hypothesis that the filtering setup expands every subnet to at least /24, the same was done for the visualization and the result is shown in Figure 5.4.

As visible in that Figure, almost all uDTAs are covered this way. The expansion also does not cover a single ATA that was previously colored black, so it is not over correcting. Two /24 subnets in the top left corner are still red but these are known as internal test nets of CAIDA and are therefore not available to the NT. When this “explanation” was given at a meeting with CAIDA, no direct confirmation was provided. Investigating the source code of the multicast system used¹, a filter map is initialized at /24 net resolution. This confirms that the telescope does in fact filter at a /24 resolution.

The amount of additional data sources and investigations necessary to produce an accurate map of ATAs shows how outdated and underdocumented the filtering implementation is. With the switch to the new ARDC API, almost all subnets separately announced in BGP (yellow in the figures above) were added to the filterlist used by the NT. While the expansion of subnets marked by ARDC is not publicly documented, there now exists awareness for it.

¹<https://github.com/CAIDA/dagmulticaster/blob/master/src/darkfilter.c>

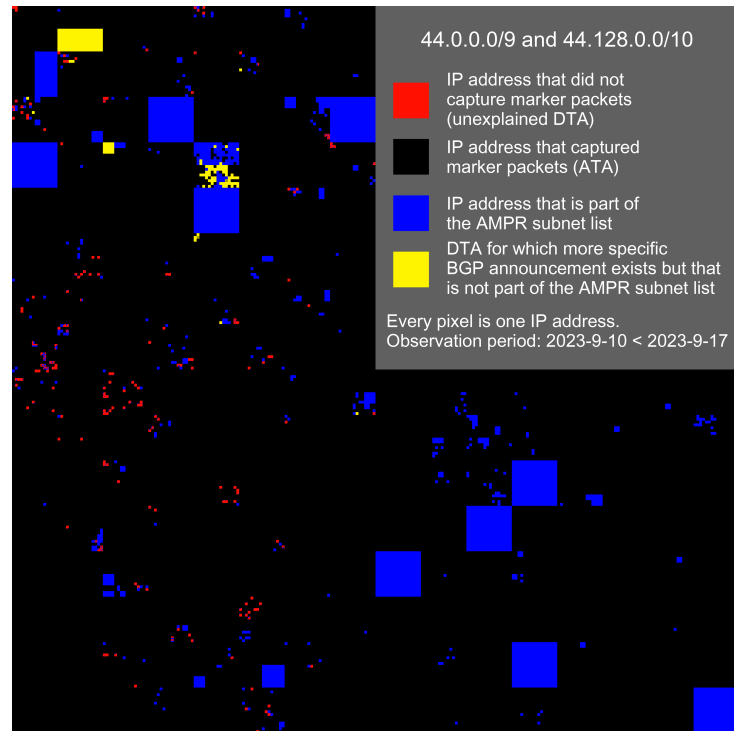


Figure 5.3 Hilbert curve of AMPRnet, with uDTAs covered by more specific BGP announcements colored yellow.

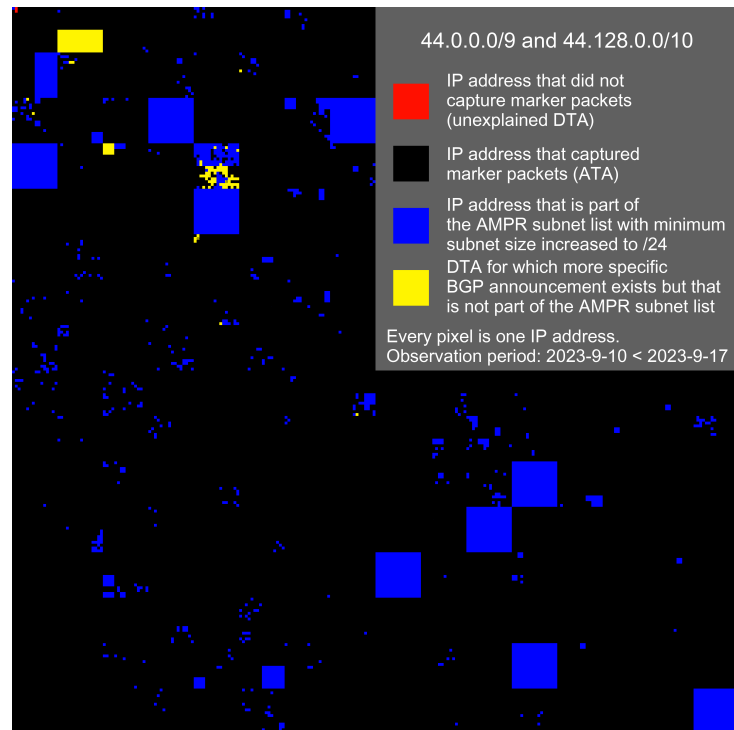


Figure 5.4 Hilbert curve of AMPRnet, with addresses covered by more specific BGP announcements colored yellow and entries of the subnet list expanded to their covering /24.

5.1.5 Impact of a Strict Filterlist

For the goals of this work, it should be fairly obvious why a well behaving and known filter setup is required. If it is not provided, packets and flows might be incorrectly flagged as loss. That is even if the packet has actually reached its destination, it just not being the NT but a ARDC lease holder. But even for more common research involving the telescope a correct filter setup is paramount. When looking at traffic captured by the NT and analyzing patterns, both an overly loose or too restrictive filterlist can have a negative impact.

In the case of a filtering setup too strict, precious IP addresses which could actually be used by the telescope, are lost. This is the case with the current filtering setup expanding subnets to /24. Most ARDC address assignments only include four or eight addresses. These /30 or /29 prefixes get expanded at the filter, which in the worst case loses the NT 250 valid IP addresses per assignment. Given that almost half of the ARDC subnet list contains prefixes more specific than /24, this amounts to 41141 IP addresses or 75% of a /16 net excluded without reason. Many other network telescopes operate on less IP addresses than that.

Given the size of the CAIDA NT, the excluded IPs only cover 0.3% of the entire address space. For every bit of higher resolution, more state has to kept by the filtering software. Given the current code architecture, the state would increase from currently 131KB to 33.5MB to filter on an individual IP address level. While the pure size increase is not massive, things like caching have to be considered. Additionally changes to the filtering progress are risky and change the way comparisons to historical data can be drawn. Still, this might be a direction worth exploring in the future for the operators.

Another problem with the list provided by ARDC is that any address assignment will be blocked for the telescope. This does not include any check for if the addresses are actually used. When the switch to the new ARDC API happened a lot of new prefixes were added to the list. The impact is visible in Figure 5.5 which shows the total volume of packets received by sources belonging to SC2 in a week. The new prefixes included some /16 nets, which are now blocked completely for the NT, even though only small parts of those /16s are given out as leases. After a telescope related outage between November 15th and 20th and the introduction of the new list on November 16th, we see that the average number of packets has dropped by some 10000 per hour. The missing packets are now getting filtered out. As a result of the new filterlist, at least eight hundred thousand additional IP addresses became DTAs. This further cuts into the number of IP addresses available to the NT. Still, its better to remove too many IP addresses from the NT rather than catch non-IBR traffic incorrectly. That is both from a research as well as an ethical point of view, since otherwise legit traffic with personal information is stored in the NT dumps.

5.1.6 Impact of a Loose Filterlist

In case of the lease holders that announce their address blocks via BGP themselves, one could argue that no filtering at the NT has to happen. Since their prefixes are more specific than the prefix announced by the UCSD for the entire AMPRnet, Internet hosts will automatically route the packets away from the telescope and towards the correct recipient. An incident that occurred on September 14th 2023 shows that relying on BGP is not sufficient.

On that day, AS57853, a legitimate lease holder of a /24 block, withdrew all its BGP announced prefixes for a duration of two hours. As a result, packets sent towards that /24 were now routed to UCSD and the NT, since the broad announcement for AMPRnet was now the most specific prefix. The IP addresses contained in the leased address block were not in the filterlist used at that time. Therefore no filtering on ingress at the NT happened and packets destined for the /24 were stored in the PCAP dumps of the telescope. From these stored packets, it also became apparent why the original announcement was withdrawn.

Almost all packets received by the NT in the time frame were large DNS response packets with domains ending in “.gov”. These domains are often used for amplification DDoS attacks [33]. In this context, the withdrawal by the lease holder might have been a protective measure or just the result

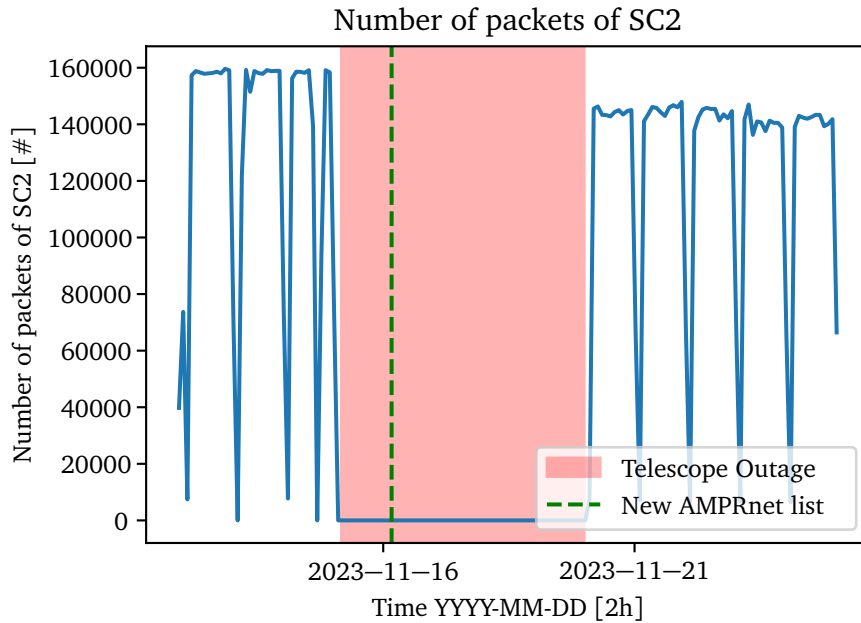


Figure 5.5 Number of packets received from SC2. The green line shows the introduction of the new filterlist.

of the router handling the BGP peering sessions for this AS being overwhelmed and connections terminated.

The exact reason for the withdrawal does not matter for the NT. What matters however, is that packet which should not reach the NT were ultimately picked up by it. Legit traffic to the lease holder might be included, or false conclusions might be drawn from the fact that the NT seemingly became the target of a reflective DDoS attack. In fact, this event was only picked up since the traffic stored in the PCAPs included some marker packets from SC1, which was running a scan in the period the prefix was not announced by the lease holder. Due to the random nature of the scans by SC1, not every IP address within the /24 block was hit. Figure 5.6 shows the resulting block in the visualizations from before. As some addresses received marker packets, they were deemed ATAs and colored black. Addresses that were not scanned during the two hour window were marked as DTAs. Since the prefix was still announced when the RIB dump used for the classification of BGP announcements was created, they were colored yellow. Would the prefix have stayed announced, the entire /24 block should be colored yellow.

This unusual pattern caused further investigation, which revealed the BGP withdrawal. If not for the scan of SC1, this event would not have been picked up by any system. Subsequently all packets received would be stored and in turn used for research, even when they do not represent the traffic usually sent towards the NT.

5.1.7 HAMnet Data Collection

Upon further investigation into the 800k IP addresses that were excluded with the introduction of the new filterlist, two /16 prefixes were of special interest. These were 44.156/16 and 44.157/16. The region is allocated to HAMnet, a sub organization within ARDC for European radio operators. While most of the IP addresses within HAMnet are not assigned to lease holders, there are quite a few leases given out. Cross referencing with the older filterlist revealed that **NO** HAMnet leases were previously filtered out. As a result, the data of the CAIDA NT contains all legit HAMnet traffic handled by the Internet-to-AMPRnet gateway for at least the last six months before November 16th, but possibly years.

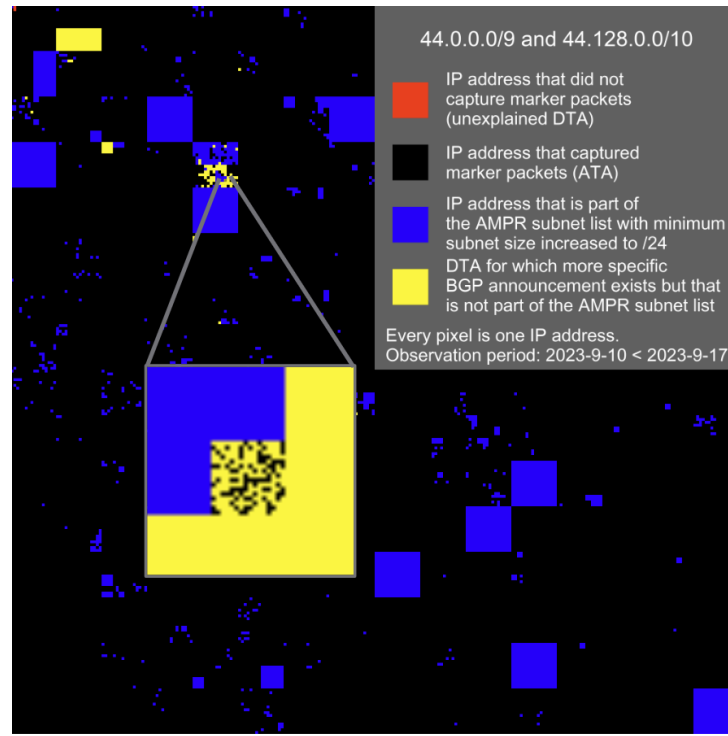


Figure 5.6 Hilbert curve of AMPRnet, with a highlighted area where only some IP addresses within a block received scanner traffic.

Within the data collected for this work, we observed a side effect from these missing HAMnet prefixes. Since the transit provider VP2 has a high exposure to European traffic, packets from HAMnet clients towards the Internet-to-AMPRnet gateway co-located with the CAIDA NT were sampled and therefore included in the analysis. The source IP address with the highest number of packets sent during the time frame was a member of HAMnet. The addresses contacted by this source only lay within the HAMnet prefix 44.156/16. A manual inspection of a sample of destination addresses showed a perfect overlap with registered² HAMnet members.

However, the protocol used by this source was not IPinIP, but rather these were non encapsulated packets. The operator of this source likely did not correctly configure his device. As a result, these packets were not sent as IPinIP wrapped packets. Since all destination IP addresses were not on the filterlist, the telescope recorded all these packets as normal IBR.

Table 5.1 IP Protocol distribution of average IBR (CAIDA NT wide statistic) and the observed misconfigured HAMnet Source.

Protocol	Average IBR	HAMnet Source
TCP	93.82%	16.32%
UDP	5.01%	65.39%
ICMP	1.13%	18.29%
Other	0.04%	0.00%

It provides a perfect example for how non-IBR traffic can interfere with research. The source in question sent over 10 000 packets per hour and makes up 52.7 % of all packets considered in the analysis of VP2s traffic. If not excluded manually, this would have skewed observed distributions and behaviors significantly. Table 5.1 shows this. The usage patterns of legit hosts differ greatly from the typical TCP SYN packets received by the NT. UDP is the predominant protocol for the HAMnet source,

²<https://hamnetdb.net/?m=subnet&showAllLines=1>

carrying layer four protocols for route discoveries. Signals like these could be appealing to researchers, as the deviation from average traffic is quite large. Without having operational knowledge over the CAIDA NT, there is no reason to expect wrong data within the NT pcap dumps. Excluding them from the dataset should therefore be a high priority task. It also highlights the importance of keeping the filterlist up to date, as even excluding all IPinIP traffic would have not been enough to remove this non-IBR. Given the complex setup of the AMPRnet overlay network, cases like these are bound to happen.

It is also likely that other work utilizing telescope data in the past is affected. The impact depends on the exact research question and data selection pipelines.

5.2 Network Telescope Under High Loads

The capturing machine of the NT (“Hubble”) has been continuously running for over five years, in which the hardware setup was not upgraded. In the same timeframe both the volume and the number of packets per second received by the NT have increased by more than 50%. During traffic peaks, this can lead to overload situations. Plans for expansion and updated hardware have already been made by CAIDA, but until it is deployed, users of the NT have to cope with the issues arising.

5.2.1 Broken Captures

Under peak conditions, the capturing machine will some times run out of storage and RAM, as the traffic amount received in an hour exceeds available disk storage. As a result, PCAP files can be cut-off early or might be missing entirely depending on how fast recovery of the capturing process happens. This then also applies to the FlowTuple data source, as they are created from the captured PCAPs.

While validating the data processing pipeline, multiple of these cases were observed within the observation time frames. Before sanitizing for these capture failures, the data processing pipeline reported 100% packet loss during these events.

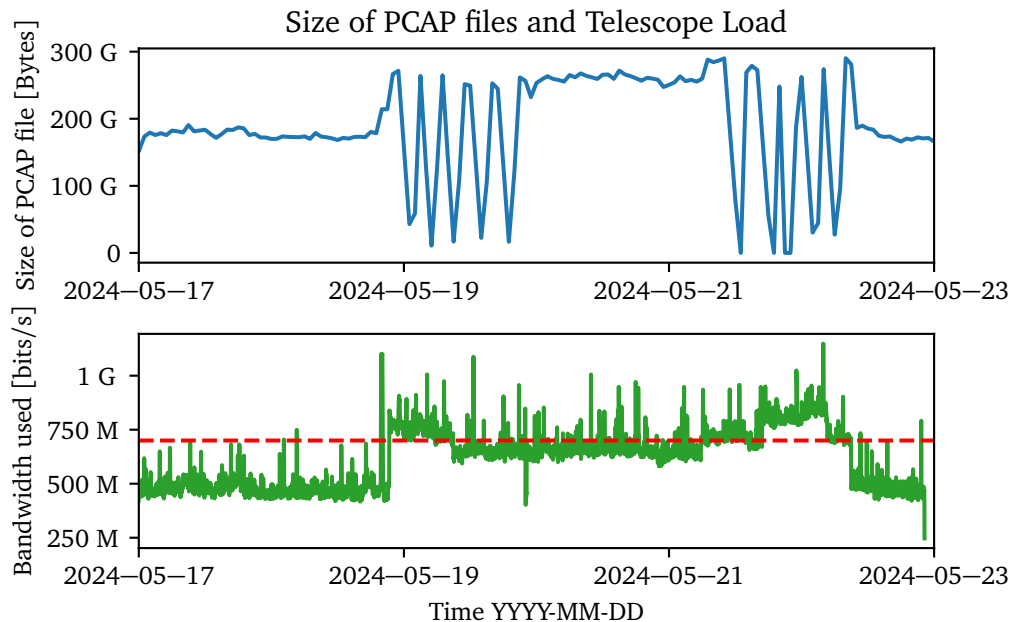


Figure 5.7 Size of stored PCAPs files of NT and bandwidth utilization.

CAIDA was already aware that this was happening, but no central monitoring or status page existed. Any larger accumulation of lost packets had to be manually inspected and stored PCAP dumps verified.

While completely missing files were easily detectable due to the known rotation period of the dump files, corrupted or cut-off captures proved more difficult. File size showed to be a decent metric for determining the status of a capture. On average, a compressed PCAP file storing one hour of captured packets would have a file size of 170GB. Even in hours of low activity, the file size would never drop below 100GB. Figure 5.7 shows the size of stored PCAP files in relation to the volume of data received by the NT. In case of sustained loads above 700 Mbits/s, storage capacity can be exceeded and as a result capturing temporarily breaks. The system recovers on its own after a certain time.

Tracking the size of captures files therefore provides an indicator for capturing issues. If storage capacity is exceeded near the end of an hour or capturing is recovered in the first minutes of an hour, inspecting file size may not suffice. Outages below 15 minutes usually do not get detected by file size alone. For this, manual inspection of each capture file is still required. Since the telescope receives on average roughly a million packets per second, it is extremely unlikely that no packet in the first or last second of a time period was received. The timestamp of the first and last packet can therefore show whether capturing stopped. The utility “capinfos” provided by Wireshark helps extract these timestamps for verification. There still exist cases however, where both these checks might overlook a capturing error. If the outage occurs within a single period and is short enough so file size is not impacted heavily, the outage remains undetected.

For validating the data used for the following analysis, a monitoring system was developed. It periodically pulls file sizes for both the PCAP data source as well as the FlowTuple data source. If a file is missing or below a certain threshold size, an alert is issued. Additionally, historical data is stored and provided in a dashboard, to crosscheck older observation periods. This monitoring system has been provided to CAIDA and the alerts issued are now routed into internal communication systems. Since its inception, it has already detected multiple outages and correctly notified CAIDA staff. Its also been tracking ongoing issues with the FlowTuple data source.

5.2.2 Dropping Packets

While the issue above causes total loss of data for a certain time frame, the NT suffers from partial packet loss as well. This also happens under peak load events. The cause for the partial packet loss is not known currently. It happens either if the packets per second count hits values of above 2.5 million, or total utilized bandwidth is above 1.2Gb/s. Both these thresholds are approximate, with the length of the excursion also seemingly playing a role.

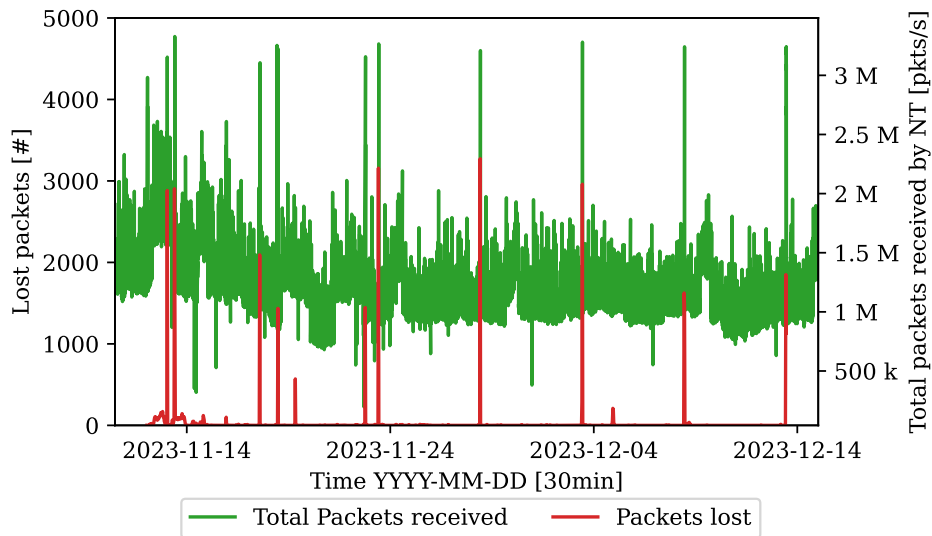


Figure 5.8 Measurements of SC2 with packet drop in relation to total packets received by NT.

The packet loss first became apparent when investigating spikes in missing packets from SC2.

Figure 5.8 shows the number of packets lost from measurements. High peaks, usually spanning no more than 20 minutes each, are visible. In each peak, all sources of SC2 are impacted equally, which indicates issues very close to the telescope and not with individual upstream providers. The packet loss is uniformly distributed within every single measurement, so all TTL values are impacted. The number of packets lost per measurement appears to be normally distributed, with many measurements missing one to three packets, but up to five lost packets ($\sim 45\%$ loss) in a single measurement were observed.

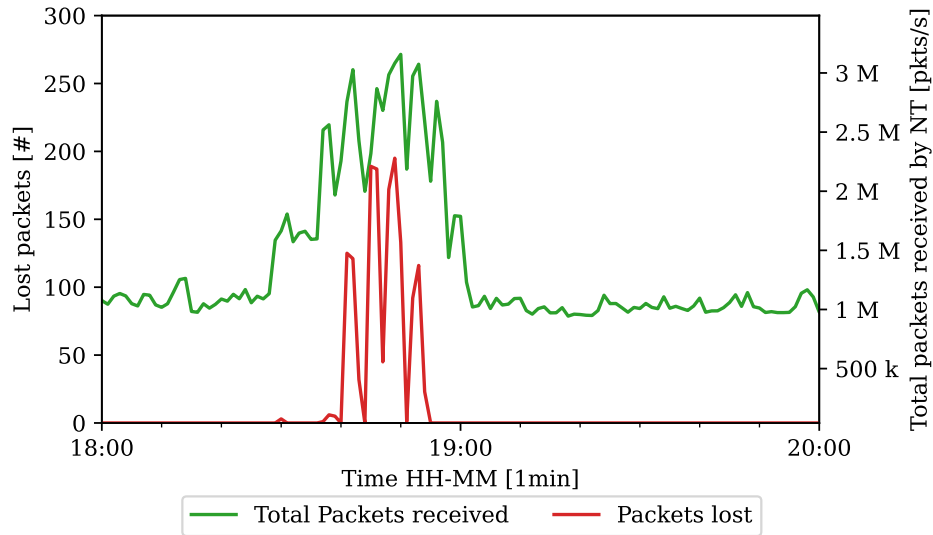


Figure 5.9 Measurements of SC2 with packet drop in relation to total packets received by NT on Nov 22nd

Figure 5.9 shows one such spike, which occurred on Nov 22nd at 18:40 UTC. We can see the typical length of an event, as well as how accurate the lost packets of SC2 track the duration of the event. During this exceptional load, 9.2% packet loss for SC2 was observed. Given that there is no external reason to assume that packets of SC2 are more likely to be dropped than others, the telescope as a whole probably experienced roughly 9% packet loss. As these events happen relatively frequently, with traffic excursions appearing at least once a week, further investigations are required.

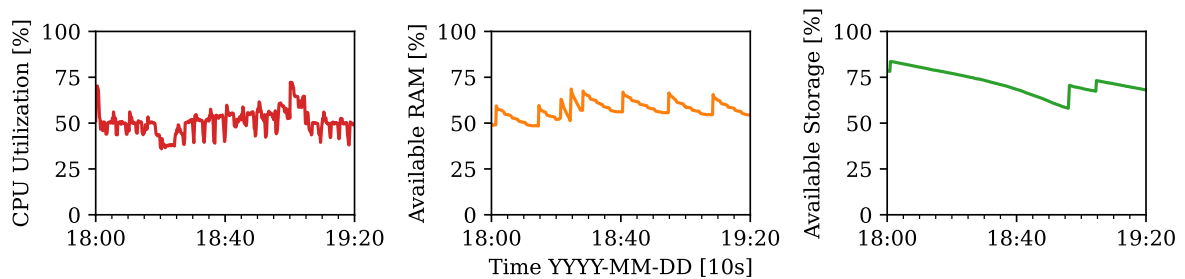


Figure 5.10 Resource Utilization of NT capture machine on Nov 22nd

At the time the existence of this issue was raised to CAIDA, they were not aware of this happening. In contrast to the completely broken capture files, these events do not show up in monitoring that only looks at CPU, RAM and storage usage. In fact, when looking only at hardware utilization, everything looks fine as Figure 5.10 shows.

The limitation in this case probably lies within the communication between the network interface card and the capturing software. Many small packets can overwhelm certain queues and storing layers on the path from the network card through the system kernel to the capturing software. Some

of these buffers are ring buffers, where a single packet, regardless of its size, takes up a slot. Very high packet count can therefore expose bottlenecks that pure high throughput might not. The exact issue is not tracked down yet and the investigation together with CAIDA is ongoing.

Given that these events happen on a regular basis, they cannot be easily glossed over. In fact, some of the spikes in Figure 5.8 are likely caused by the same scanning origin in a weekly pattern. The telescope loses a percentage of total traffic during such an event, not just the packets from the sources causing it. This impacts all analysis done in the time frame, as background traffic decreases by the percentage, without any change in real world conditions. If a port or protocol is not part of the spike traffic, it too incorrectly experiences a reduction of observed packets in the telescope data. As a result, classifications or long term studies can be impacted by patterns in stored data not actually present in IBR.

6 Results

The original goal of this work was to gain an understanding of potential blindspots of the NT due to middleboxes and other filters. A lot of time was dedicated to account for the operational issues of the telescope, as this was a requirement for any further analysis. Even though not all issues with the NT could be resolved, with the insights gained, the data can be sanitized and the original research question be studied.

6.1 Connectivity-Related

In case of the following two findings, their base causes lie in BGP and with the connectivity setup of the telescope. With ASes and networks having different internal layouts and connections to the telescope and its upstream providers, problems can arise. During the time frame this work was performed, an issue with BGP filtering was observed. The data source SC2, which operates a setup with uniform sources, showed the impact of shorter and longer paths on measurements.

6.1.1 Visibility of Chinatel Customers

During the analysis of packets received by SC2, one source was noticeably absent. The upstream provider for this node is “Chinatel”, a chinese ISP. The operator of SC2 also reported that traceroutes from that node towards the telescope were not reaching beyond hops of Chinatel. Other measurements to more specific prefixes within AMPRnet, announced by different ASes, were not impacted. While initial theories included censorship measures towards an IP range announced by an American university, this was not the case.

After further investigations by the operator of SC2, they noticed that no prefixes routed via AS3356 (Level3) could be reached by the Chinatel source. Most more specific prefixes were routed via another Tier-One transit provider and were therefore not impacted. Raising this issue to Chinatel, a two week debugging process yielded the discovery of a filtering problem with their BGP routers.

As a result, Chinatel did not propagate announcements of the SC2 operator to over half of their Tier-One peers. This list included Level3, which resulted in them dropping ingressing packets coming from SC2 controlled prefixes via Chinatel. Since SC2 is a normal customer to Chinatel, there is no reason to suspect any special treatment. Until this problem was resolved in November 2023, likely no customer traffic handled by Chinatel was able to reach the NT. This creates a considerable blindspot, since Chinatel is a big provider in the region.

According to CAIDA AS Rank [3], their AS4134 customer cone spans 471 ASes, with over 300k IP addresses. While not all of their customers have to select the routes provided to them by Chinatel, the fact that many of their customers only have Chinatel as an upstream means that most customers do. The NT therefore had a severe blindspot for Chinese traffic until the issue was fixed. No time

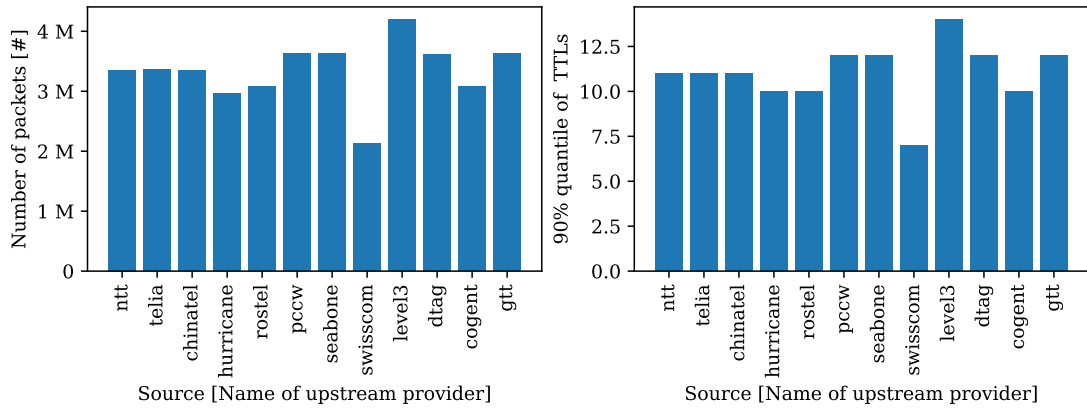


Figure 6.1 Number of packets received by each source of SC2 and the top 90% of TTLs by each source.

is known for when the propagation stopped, so the full extent is not known as well. In the CAIDA Flowtuple source, derived from the pcap dumps and stored for longer timeframes, the IP address of SC2s Chinatel source was last seen in March and April of 2023 but is missing from December 2022 up to that point as well.

6.1.2 Impact of Hops to the Network Telescope

The telescope is hosted at UCSD and its prefix announced via the UCSDs AS7377. This AS is connected in IPv4 via the regional educational network CENIC (AS2152), which has providers Level3 and Hurricane Electric. Most of the traffic comes in through Level3. The peering point is in Los Angeles.

```

1 ae2.3602.ear2.LosAngeles1.level3.net
2 CENIC.ear1.LosAngeles1.Level3.net
3 dc-lax-agg8-lax-agg10-300g.cenic.net
4 tust1-agg-01-lax-agg8-400g-01.cenic.net
5 sand1-agg-01-tust1-agg-01-300g-01.cenic.net
6 ucsd-sand1-agg-01-100g-01.cenic.net
7 nodem-core-6807-vlan2767-gw.ucsd.edu
8 sdsc-7710-7-mcore-vl2995-p2p.ucsd.edu
9 gw.ampr.org

```

Listing 6.1 List of hops every packet incoming via Lumen takes to the NT.

Listing 6.1 shows this from a hop list perspective. After a packet has reached the Level3 router in Los Angeles, it still has to traverse 8 different hops before reaching the telescope. For packets which already have a low TTL, this might result in a drop before they reach the NT.

This is especially true in the case of traceroute measurements like those of SC2. Figure 6.1 shows the total number of packets received from each source, categorized by their upstream providers. Even though each source performs the exact same traceroute measurements with the same maximum TTL value, we can see differences of almost 50% between the node connected via Swisscom versus the node sending its packets to Level3.

As shown in Figure 6.1, the highest TTLs received by the Swisscom node is halve of the value of the Level3 source. While the system uplinked with Level3 has the shortest path to the telescope and therefore the most packets that reach it before the maximum TTL packet is sent, Swisscom has the longest path. In their case, they send traffic through Hurricane towards the NT, a provider that already has a slightly longer path than others. Topological distance to the NT therefore matters for some applications.

All of these nodes are located close to their providers and have a maximum TTL of 25. In the case of Swisscom there are 18 hops between the source and the NT.

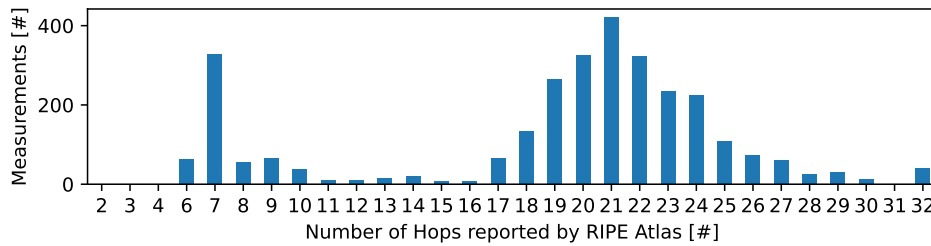


Figure 6.2 Number of packets received by each source of SC2 and the average TTL by each source.

The impact is also visible in the RIPE Atlas measurements. Figure 6.2 shows the number of hops taken in each single measurement. Every run with a reported number of hops larger than 27 will have less packets stored in the telescope, which, if not corrected for, would lead to falsely classified packet loss. The probes with these high hop counts were located in large internal networks with often as much as 10 hops via private IP addresses.

This does not impact the larger share of packets received by the NT, but might influence research working with traceroutes. The packet matching for traffic sampled by VP2 had to account for traceroutes, which demonstrates that the NT does receive this type of traffic on a regular basis.

6.2 Filtering-Related

The measurement setup with data from VP2 is not optimal to spot the interference of middleboxes, as the collection point is located after most of those would be expected on a packet path. For this reason, the RIPE Atlas measurements were of great value, as these are located in end user networks and traverse these middleboxes.

6.2.1 IP Fragmentation Losses

One of the measurements performed with RIPE Atlas was a ping with a packet larger than 1500 bytes. This exceeds the usual MTU of most networks and leads to fragmentation. For security and performance reasons, fragmented packets are sometimes dropped by routers. The RIPE Atlas data reflects this.

Table 6.1 Packet loss from RIPE Atlas probes for: normal pings, first fragmented packet, second fragmented packet

Packet	Relative Packet Loss	Relative Factor
Normal Ping	1.7%	1.0
1st fragmented packet	5.7%	3.3
2nd fragmented packet	12.7%	7.4

Table 6.1 shows the packet loss for a normal reference ping measurement using packets with a size of 64 Bytes. The packet loss includes probes that could not reach the telescope for other reasons, like no Internet connectivity. So the 1.7% packet loss are a baseline. As we can see, fragmented packets experience more packet loss. For the first resulting packet after fragmentation, loss grows to 5.7%. Of the second packet, 12.7% or a 7.4 times increase over baseline do not reach the telescope.

A potential reason for this might be that the second packet cannot contain possible Layer 4 headers (even though there are non in this case). Middleboxes like firewalls might be more likely to drop such a packet, as they are unable to perform deep packet inspection to enforce firewall rules.

As a result, the telescope probably misses out on some fragmented packets that are dropped on the path. For some use cases of the telescope this is relevant, as backscatter of attacked DNS server may be fragmented because of large DNS responses.

6.2.2 Bogon Source IP Filtering

IP addresses specified in RFC 1918 [31] are allocated for private networks. As a result, prefixes named in the RFC should not appear in the global routing table. These addresses are often referred to as “Bogons”. Since there is no legitimate reason for packets containing these IP addresses to leave a local network, operators may choose to drop them. The only reasons for why such packets do reach the Internet are misconfigurations or malicious intent like spoofing.

Spoofed traffic is potentially interesting for researchers using the NT. The transit provider VP2 did collect flows at its ingress edge routers containing Bogon source addresses. This indicates that at least some of their customers do not properly filter bogon source packets out. Non of these flows however reached the telescope, indicating that either VP2 or its peer Level3 perform Bogon filtering. Overall, 46.04% of all flows collected by VP2 that did not reach the NT had a Bogon source IP address.

6.2.3 UCSD Campus Network Filtering

The AS announcing the telescope IP prefix is that of the UCSD. Therefore, UCSD routers handle and process traffic destined for the telescope. In one of the meetings with CAIDA, a list of IP prefixes was provided. All networks contained in it are filtered by the UCSDs Campus Network, through which telescope traffic travels towards the capturing machines at CAIDA.

Table 6.2 Filterlist of UCSD campus network

Prefix	AS Holder
55.0.0.0/8	DoD Network Information Center (USA)
66.0.0.0/8	No single owner
185.176.27.0/24	OKB PROGRESS LLC (Russia)
176.113.115.0/24	Cat Technologies Co. Limited (Hong Kong)
194.31.244.0/24	LUMINA LLC (Ukraine)
195.54.166.0/24	RocketConnect OU (Estonia)
92.63.196.0/24	IP Volume Inc (Seychelles)

Table 6.2 shows the entire list. All of the smaller prefixes on the list belong to different hosting providers. It is not certain that these were also the owners at the time the respective prefix was added to this list. In case of IP Volume Inc, the company is known for hosting command and control servers of botnets [6] and is considered a “bullet proof hoster”. This category of hosters is generally connected to illegal activities, with operations often located in countries with weak jurisdictions.

No reliable information can be found for the other /24 prefixes on the list, but it can be assumed that they were added for similar reasons as the prefix of IP Volume Inc. It is not unlikely that some servers of these providers were utilized for malicious activities, substantiating their addition to the filterlist.

More interesting and impactful in terms of lost exposure for the NT are the two /8 prefixes. For 55/8, the organization associated with it is the Network Information Center (NIC) of the American Department of Defense. One can only speculate why this prefix made the list.

66/8 is a network allocated to ARIN, with a lot of more specific prefixes announced, as they were given out by the RIR. It is unclear how this network made it onto the list, but it creates a real blindspot for the telescope. A /8 is 1/256 (0.3%) of the entire IPv4 address space and 0.45% of the publicly routable address space. Dropping all traffic originating from a /8 unnecessarily limits the telescope.

Since this is not one of the big DoD /8s but one filled with small allocations from a RIR, it is a possible originator of the traffic the telescope is interested in.

The RIPE Atlas measurement campaign also coincidentally confirmed that these filters are applied to the telescope. Two of the randomly selected probes held IP addresses in 66/8. In every single measurement, the probes were able to reach the last CENIC router (see Listing 6.1), but never the first UCSD router. Even in the very unlikely event that for every measurement from these probes, responses of both UCSD routers fell victim to ICMP rate limiting, at least four packets could be expected in the NT PCAP dumps per measurement. This includes one with a starting TTL value of 255 (see section 4.3.4). Not a single packet from those source IP addresses could be found in the PCAP file for the time frame in which the RIPE Atlas campaign was conducted. There is no reason to suspect that any other part of 66.0.0.0/8 is treated differently, so the telescope has had this blind spot for the entirety of the campus filterlists existence in this form.

6.2.4 On-Path Filtering

The RIPE Atlas probes performed a multitude of measurements using different protocols and destination ports. When individual probes are able to reach the NT in one measurement, but not in another, it indicates the existence of a firewall on the path.

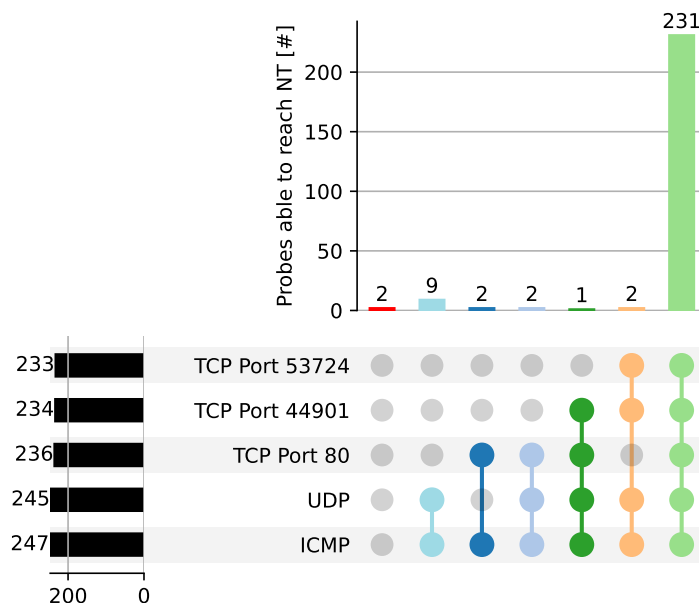


Figure 6.3 Different combinations of measurements for which probes were able to reach the NT.

Figure 6.3 is an Upset plot. It visualizes intersections between different sets. In this case, the sets contain the probes that could reach the NT in a specific measurement at least once in three tries. The graph in the bottom left shows the number of probes in each set.

In total, 249 probes participated in the measurements. Of those, only two did not reach the telescope at all. These are the two probes which had IP addresses from the filtered out 66.0.0.0/8 prefix.

Nine probes were unable to reach the NT at all with TCP. There are also four probes which either can only reach it via port 80, which is usually occupied by HTTP servers, or can reach everything but port 80.

Only two probes which were able to reach the NT at least in one measurement, could not reach it via UDP. ICMP, as far as the data shows, is never filtered out.

With 231 probes, most were able to reach the NT in each measurement at least once. These 92.7% of tested nodes likely have no firewall with filters for these protocols configured on their path. It is possible that a stateful firewall might allow the first connection but drop successive attempts. The test setup is however not designed to reliably test this.

6.3 General Packet Loss

Besides misconfigurations and middle boxes filtering out traffic, packets can become lost from overload situations or other anomalies. The next two sections contain results for this “general” packet loss, which cannot be attributed to a specific reason. Low amounts of packet loss can occur to any traffic traversing the Internet. This is not unique to packets destined for a network telescope.

6.3.1 Low Packet Loss in Tier-One Networks

Overall incomplete flows observed from VP2 to the telescope were 0.048% after sanitation.

This means that a packet was picked up at an ingress router of VP2, creating a logged flow, but no corresponding packet was found in the NT. Considering that the path taken by any packet after these ingress routers only traverses the systems of Tier-One providers and the UCSD, a low number was to be expected.

Sanitation included the removal of incomplete flows during NT outages, flows identified as UDP traceroutes and those containing bogon source IP addresses. Non the less, the loss percentage likely still contains some incomplete flows caused by NT overload. Correctly and confidently identifying these often very short partial interruptions is hard. The already low number should therefore be interpreted more as an upper bound. Without removing the partial outages of the telescope, loss is 0.08%. This 80% difference shows the importance of correctly monitoring the telescope.

6.3.2 Packet Loss due to Link Overload

The detected packet loss of SC2 after the removal of NT outages is 0.019%. Without removing the time ranges in which the NT experienced issues, packet loss would have been at 0.09%, an 4.5 times increase.

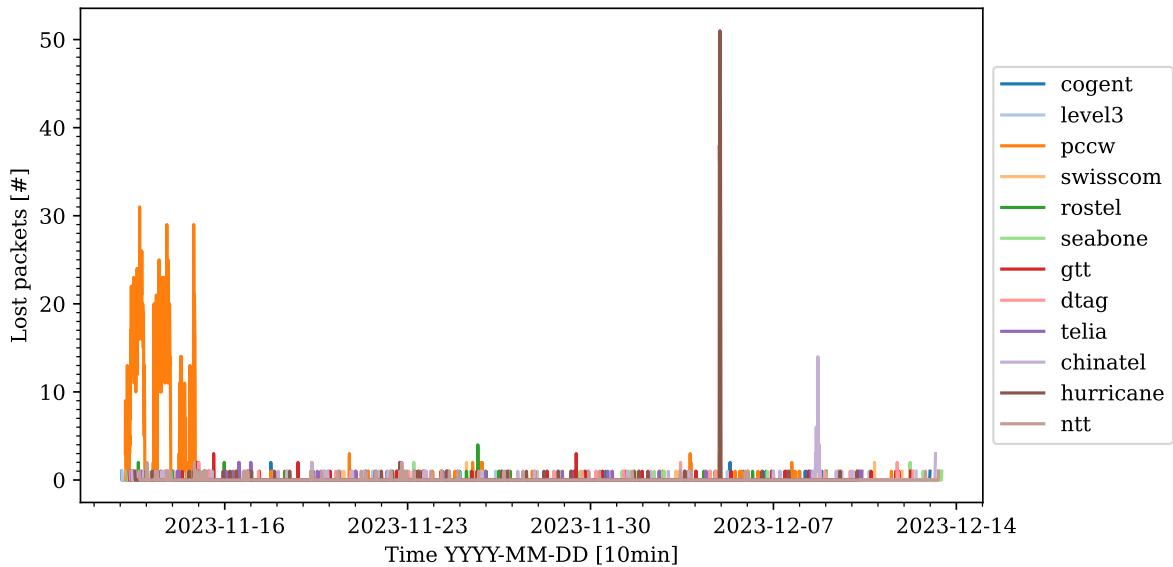
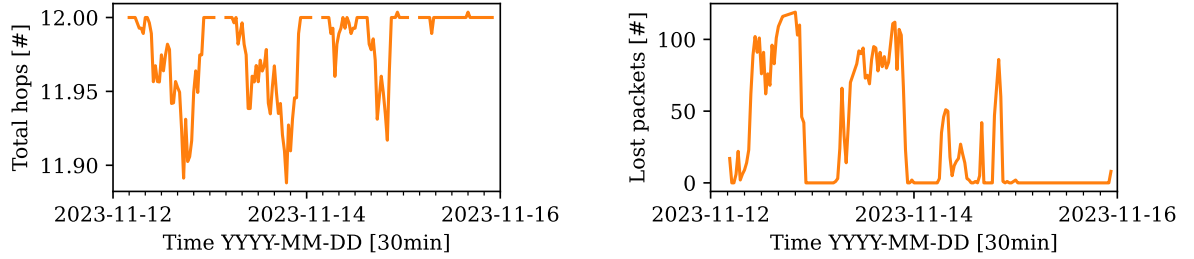


Figure 6.4 Packet loss for each source of SC2 during the observation period.

Much of the packet loss suffered by sources of SC2 can be attributed to two different events. As Figure 6.4 shows, from November 12th to November 15th, right at the beginning of the measurements, the source with PCCW as its upstream provider experienced consistent packet loss.

The time of occurrence followed a diurnal rhythm for European time zones, indicating that the reason might be a router getting overloaded during higher local traffic hours. Even though PCCW is a Chinese provider, handover for traffic coming from SC2 happens in Frankfurt am Main (Germany). A load pattern starting at six in the morning UTC and ending at ten in the evening UTC makes sense in this case.

After November 15th, no more excess packet loss events happen for the PCCW source, indicating that a fix was implemented. The data available does not allow to narrow down the applied solution.



a) Total hops observed by SC2

b) Lost packets at NT

Figure 6.5 Packet loss event of PCCW source of SC2 from November 12th to 15th 2023.

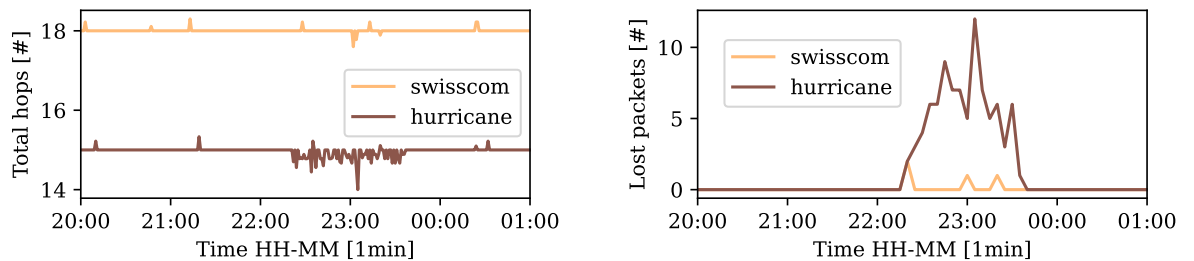
Figure 6.5 shows a correlation between the observed packet loss and the average number of total hops reported by the traceroute running on the source of SC2.

The number of total hops on the path shows some disturbance during the time of the event. An explanation for the seemingly small change in the average is that the value only drops if the packet destined for the last router in front of the telescope is dropped. Only in this case does the traceroute tool report a shorter path, since the last router before the NT is the last responsive hop.

Due to the telescope not answering with TTL EXCEEDED and some routers implementing ICMP rate limiting, the number of invalid hops was not very reliable as a metric and is therefore not represented here.

After November 14th, packet loss from the PCCW source returns to the global average. The reason for the outage and the potential ramifications done by the provider are unknown.

The second large observed event was a one and a half hour long disruption for Hurricane Electric on the 4th of December 2023. This case is interesting, because Hurricane provides the route towards the telescope for Swisscom as well. Depending on the location of the issue, both providers could be affected.



a) Total hops observed by SC2

b) Lost packets at NT

Figure 6.6 Packet loss event of Hurricane source of SC2 on December 4th 2023.

As we can see in Figure 6.6, the number of total hops again is influenced. The higher amount of

total hops for Swisscom is an indicator for a longer path with more devices on the way to the telescope. Interestingly, there are only few lost packets for Swisscom, but a lot for Hurricane Electric. This could mean, that the overloaded or failing node does not lay on the path through which Hurricane moves Swisscom traffic to the NT.

Alternatively, Hurricane might choose to prioritize Swisscom traffic over other customers, to adhere to different service level agreements. The operator of SC2 does not have any such agreements in place, so traffic from SC2 would likely have the lowest priority.

7 Discussion

Properly gauging the amount of packets getting lost from vantage points to the telescope required that any packet not being stored in the PCAP dumps was truly lost on the path. With every new individual lost packet inspected, more issues became apparent.

Since active leases in AMPRnet get more traffic than IP addresses not running anything, most traffic observed at the VPs towards the covering prefixes 44/9 and 44.128/10 was destined for leased out addresses. When such a lease is separately announced in BGP but not added to the filterlist, false positives are created. As packets sent towards the lease holder that are seen by the VPs are wrongly interpreted as lost. Hence a deeper investigation of the filterlist was required, before the main research question could be answered. This revealed a collection of operational issues with the CAIDA NT.

With the CAIDA NT being located in AMPRnet, it is of the utmost importance that the telescope stays on top of changes in the address space. For over six months however, no new leases or other changes were fed into the filterlist. This included both unnecessary blind spots, as well as actual active areas, with their traffic being falsely considered IBR. While one might not expect a lot of changes to that list, reality shows them happening at least once a day. Larger movements are also possible due to outside reasons. For example, the sale of 44.192.0.0/10 to AWS required the transfer of all leases from that address space into the remainder managed by ARDC. This included some HAMnet ranges, which may have not been readadded to filterlist once assigned again.

Current monitoring and incidents like those stated in Chapter 5.1.6 also highlight that even if a lease holder announces their prefix via BGP, it has to be added to the filterlist. Members of ARDC run experimental networks by their very nature. Often, small networks are announced for short times like one or two weeks, before they are withdrawn again.

Besides the prefixes that are announced via BGP, a large part of the leased out prefixes is only used via the overlay network. Using and maintaining the filterlist is the only way to remove their traffic from the telescope. This work has shown some high-level effects of both having non-IBR sources in the NT data collection, as well as the repercussions of large changes to the filterlist.

For longer periods of time, large parts of HAMnet were not excluded. Papers utilizing data from the NT in that time frame have been published without knowledge of this. The effect that this may have had on past research depends on the kind of study that was performed. As most ARDC members use tunneled traffic and protocols based on UDP, the impact on studies looking at TCP traffic should be low. Any work that tries to analyze UDP or ICMP traffic however might be impacted more, especially if they find only few sources for their observed behaviors.

Besides issues with the address space, the telescope suffers from complete and partial recording outages. During a complete outage, nothing is recorded and therefore can not be analyzed. The data is also unrecoverable after such a failure. Depending on the level of aggregation performed by a researcher, full outages are easy to detect. Still, they require a thorough investigation to remove

the possibility of any other part of the data processing pipeline failing. In the process they take up valuable time of researchers. Even after full outages are accounted for, they create holes in an otherwise continuous observation time frame.

Something to keep in mind with both full and partial outages is that they do not occur randomly. Rather, they are a result of higher than usual traffic volumes. These high volume events are some of the most interesting to observe. In these critical moments, the NT falters most often. This gets even more insidious with the partial packet loss.

If a research project does not involve any ground truth traffic, there is no way to automatically detect partial loss. It then can not be accounted for and if observation times are short enough, significantly skew statistics and aggregations. Every time the telescope is hit by more than three million packets per second, even the time series directly published by CAIDA are impacted. It shows a drop in both packets per second and the number of unique IP addresses for source countries not involved in the event causing traffic. The drops both match the duration of the event and the observed packet loss of ten to twenty percent.

Past [17] and present research efforts try to use methods for detecting variations in time series to classify types of events. These projects can receive false signals from derivations caused not by real world traffic but telescope interference. In conversations with researchers at the GMI-AIMS workshop hosted by CAIDA, this was confirmed to have happened during the development of a new anomaly detector. Impact will vary between different research projects.

Until November 2023, there was no dedicated monitoring for any telescope-health related metrics. As part of this ongoing work, reoccurring validation of the file sizes was implemented at the Hackathon of IETF 118. The system has since then alerted operators multiple times on both short and long term issues with the stored data sources. Beginning in July 2024, partial packet loss is being monitored. This was implemented during a visit to the GMI-AIMS 3 workshop. Using a detection method similar to how the partial packet loss was recorded in this work, it will hopefully allow a deeper understanding and help find possible fixes for these problems.

These issues do not just apply to the NT run by CAIDA but rather these are general challenges with operating a telescope of this size. With ever increasing traffic numbers, constant attention and upgrades are required to keep the system in good shape. Understanding and staying up to date with telescope health also differs a lot from just monitoring the system statistics of the host machine. The telescope requires constant active maintenance. From the view of a researcher, this is a lot of time not spent on producing results, and so it is seldom a high priority.

With regards to the address space and the resulting issues from it, there is not really any alternative. The state of IPv4 address exhaustion requires compromises like using AMPRnet. For public research, acquiring over 200 million dollars worth of IP space would be impossible. The cooperation with ARDC again makes active and ongoing maintenance activities mandatory. The dynamic nature of AMPRnet creates a lot of complexity. It gets amplified by the fact that CAIDA relies on the cooperation of an entity run only by volunteers. ARDC staff do understandably not have infinite time to spend. The telescope is also not the main purpose of the address space and its managing organization. As a result, its often an afterthought to any changes done by ARDC. An example would be the deprecation of APIs in active use by the NT without any notification.

In a perfect world all these issues could be solved by owning IP space, but this is unrealistic. With CAIDA focusing more on upgrading and monitoring the telescope in recent times, the conditions are improving.

A big issue with the current setup of the telescope is that it is located behind the campus network filter of UCSD. The reason for why this is the case is not known. The existence of such a filter makes sense for the university, to block IP ranges known for hosting malicious activities. For the telescope these are the most interesting IP ranges and packets. It creates a real blind spot not just for spoofed traffic but also real malicious sources. With two RIPE Atlas probes using IP addresses in 66.0.0.0/8 by chance during the measurement campaign, the application of the campus filterlist was also confirmed. The telescope is completely blind for 1/128 of the entire Internet. 55/8 might not be very active but

66/8 is used for example by ISPs and therefore of high interest to the telescope.

This work originally set out to search for potential blind spots of network telescopes caused by middleboxes. The CAIDA NT acted as the real world testing ground. For this, multiple data sources were utilized. Each source provided a different position and a specific view on the path that every single packet takes.

Most devices performing firewall tasks and filtering are located close to the source device. For that reason, VP1 and VP2 may not be the perfect observation points to answer the research question. Traffic that they see has already cleared most middle boxes, at which point packet loss is generally very low.

Still, these data points were highly important for enabling other results and to understand how the telescope receives its traffic. Both vantage points only see relatively small amounts of packets destined for the NT. This however allowed for a deeper analysis of individual sources and flows.

Large parts of the traffic sampled by the IXP(>70%) and the transit provider(>50%) were actual AMPRnet traffic instead of packets with true monitored NT IP addresses as destinations. While that traffic would be hard to find in the massive amounts of data collected by CAIDA, the VPs made it more visible and in turn allowed for a better analysis of the operational problems of the telescope.

The two vantage points showed that loss was low in the latter parts of the packet path. Even with telescope outages potentially increasing the numbers, only 0.05% of packets were lost. The data gives confidence that there are no operational issues or filtering solutions other than the campus network filter in front of the telescope. Since addresses in 66.0.0.0/8 are not in the customer cone of VP2 in significant numbers, the filter has no impact on the observed packet loss. No flow observed at VP2 had a source IP address in 66.0.0.0/8.

Data from both VP2 as well as RIPE Atlas shows that some commonly known and widely used filters are applied on the path to the telescope. Bogon source IP addresses were responsible for 46% of lost flows, as either VP2 or Level3 implemented Bogon filtering. RIPE Atlas measurements showed that packet loss increases for fragmented packets. In case of later fragments, packet loss increases seven-fold.

Bogons are interesting for the telescope, as they are often the result of spoofed traffic. When attackers perform spoofing based DDoS attacks, they generate a random source IP address for each packet. Since bogon IPs cover a significant part of the IPv4 address space, they will be selected with a certain chance. Looking at bogon source IP addresses in the telescope could therefore be used to research the prevalence of spoofing. As part of backscatter, the telescope also receives fragmented packets. This for example happens if a DNS server is attacked and responses too large for a normal MTU are sent by the attacked server. The backscatter created becomes fragmented.

The telescope therefore underestimates both bogon traffic as well as the amount of fragmented traffic sent by sources. The question with these types of traffic is, if the telescope should even try to represent them more. The amount currently received represents the actual amount reaching systems connected to the Internet. It is unlikely that the telescope receives distributions differing from other IP addresses. Any effort to increase this would cause a divergence between the real world and telescope collected data. If trying to draw conclusions from the traffic stored by the NT on these types of packets, this has to be kept in mind. We would argue, that it is not a blind spot that makes the NT less representative. Instead, a network telescope is likely not right choice of tool if one aims to research this kind of traffic.

The same goes for the outages of specific providers as observed by SC2. Link saturation and hardware failures are a part of the Internet. During the observation timeframe, the main upstream provider for the telescope did not have any outages on connections to the NT. As long as there is no reoccurring outages impacting all traffic towards the telescope, these are non issues.

RIPE Atlas, with its probes partly located in end user networks, is better suited for the analysis of deployed middle boxes and their effect on traffic. Results from the measurements performed in this work showed that filtering is indeed happening.

Even though 92% of probes had unobstructed paths to the telescope, 3.6% of probes could not

reach the NT with their TCP packets. In consideration of the fact that over 80% of the traffic collected by CAIDA is TCP, any filtering of this protocol has a proportionally larger impact. The impact also depends on the distribution of sending sources. If that distribution would tend more to private households for example, it is unlikely that they have filters in place. If a majority of sources is located in corporate networks, the impact of filters is greater. Since this distribution is not known, no more concrete statements can be made.

A few probes were also unable to reach the telescope in various other combinations of ports and protocols. From the low number of probes in each category, it is hard to draw any conclusions. Both allowing only port 80 and blocking port 80 specifically makes sense in certain scenarios. A larger test, with more RIPE Atlas nodes would be needed to attain more data and confirm that filters like these do exist in higher numbers. Ports alone can also not unveil the full picture. Middleboxes may choose to drop TCP packets to port 80 if no HTTP header can be found inside the packet for example. This case cannot be determined from the current tests.

It is also important to keep in mind that even though RIPE Atlas probes are run by end users and therefore located close to them, they are still run by volunteers. The average person running a RIPE Atlas probe is likely more knowledgeable in the realm of IT and networking than the average Internet user.

Most RIPE Atlas probes are probably intentionally not deployed behind firewalls by their operators. It would be in direct conflict with the goal of the platform. The specific question it is used for in this work is not a main consideration. It is reasonable to assume that the numbers provided here are lower bounds. With many RIPE probes being operated in universities and network related research institutions and businesses, only a small percentage will end up in corporate networks with stricter filters.

For this reason, other active measurement projects like for example CAIDA Ark would be less useful. Operators of Ark nodes are for example instructed to not put any firewall restrictions on the connection of these devices. For the measurements usually performed by these projects, such a policy makes sense. It however limits the exposure to relevant devices for answering the research question of this work.

8 Conclusion and Future Work

Conclusion In this work, we took a comprehensive view on the question why traffic that should reach a network telescope was not captured. We analyzed the reasons behind and empirically quantified the amount based on data from one of the largest network telescopes deployed. We also identified traffic that reached the telescope but was intended to be forwarded to other networks.

We mastered several challenges. To establish ground truth in the amount of traffic that a telescope should have received, we conducted active and passive measurements. Active measurements explicitly targeted addresses of the network telescope. To keep the measurement footprint small, we utilized existing scanner projects in addition to our own active measurements via RIPE Atlas. Our passive measurements have been collected at a large regional Internet Exchange Point and in the network of a Tier-One Provider. The vantage points collected IP packets destined to the telescope. These datasets allowed us to compare what was sent to the telescope and what the telescope captured.

Our method allowed us to discover the existence of both on-path and operational challenges faced by the CAIDA network telescope. We showed that on-path filtering of specific protocols occurs, but is not significant. Packets commonly known to be filtered out by operators, *i.e.*, packets with bogon source addresses and fragmented packets, contribute most to the traffic that does not reach the CAIDA telescope. Finally, on the data plane, we found that the firewall protecting the telescope introduces limitations that lead to unwanted packet filters.

On the control plane, we identified three core problems due to BGP misconfigurations. First, parts of the telescope address space were incorrectly announced by other autonomous systems, leading to unwanted traffic redirects and, consequently, loss of telescope traffic. Second, the telescope incorrectly received traffic, when more specific prefixes in the telescope address space legitimately announced by other autonomous systems, were unexpectedly withdrawn. Third, through incorrect BGP prefix filtering at one Chinese ISP packets destined for the telescope were not routed correctly, leading to loss of telescope traffic from that AS and its customers.

In addition to problems of the control and data plane, we found that proper systems monitoring of the telescope infrastructure is utmost important. Several outages occurred because servers were overloaded or network cards not well configured.

Our findings have relevant implications on the results of prior work. The telescope is generally considered to have a fixed number of IP addresses it listens on, an assumption we showed does not hold, specifically for long term studies. We also showed that the telescope collected non-IBR traffic, which weakens any conclusions drawn about IBR in previous related work. While impact depends heavily on the concrete research performed, the fact that these issues exist shows that the research community does not yet have a complete understanding of network telescopes as measurement instruments.

While most of these findings were made with the CAIDA telescope and some are unique to the setup this instrument operates in, it is likely other telescopes suffer from similar issues.

Future Work Our work has clearly shown that various events can cause traffic to not be captured by a network telescope. Some of the reasons relate to operational problems of the telescope infrastructure, which should be fixed, some are intrinsic to the Internet architecture, in addition to cases where sources actively avoid telescope address space. The fundamental question remains: which traffic do the packets captured by a network telescope represent? To answer this question, more detailed on-path observations, conducted both close to the edge as well as the core, are necessary and should be correlated with the traffic received by a telescope. Additionally, a parallel telescope rooted at the same upstream provider and geographical region could be used for cross checks.

Besides this, we identify five research questions worth exploring in the short and midterm in order to deepen our understanding of both the CAIDA NT as well network telescopes as an Internet measurement instrument.

Given the capture of GRE traffic by the CAIDA NT, additional investigations into its origin should be conducted. As GRE is one of the protocols potentially sent by misconfigured HAM operators, it is likely to reveal additional oversights or issues with the filtering performed in front of the telescope. Even if assuming the sources sending these packets are not ARDC members, an analysis should be considered. To the best of our knowledge, there are no attacks or exploits utilizing GRE, so its existence in telescope records is unexpected.

Over the entire time frame the CAIDA NT was operated, the address space it occupies has undergone a few major changes. With the sale to AWS by ARDC, 25% of available addresses were removed. A study of the effective size of the network telescope, for the entire period the telescope was operated in its current form, would allow to better scope the impact on previous work.

Related work [38, 50] that presents new types of telescopes, often argues, that static telescopes are avoided by scanners. This would also effect the CAIDA NT, as it has been using the same address space for over 20 years. Other telescopes have less publicized prefixes. Comparing the traffic that is recorded between different telescopes, this research question could be solved. Other work [18] has also shown that different telescopes see different traffic depending on the geographic location their IP addresses are associated with. A comprehensive comparison between multiple different telescopes would allow to quantify these differences and help answering whether multiple telescopes are necessary for a complete understanding of IBR on the Internet.

Active measurements via the RIPE Atlas platform performed in this work showed that middleboxes on path influence the traffic heading towards the CAIDA NT. To bolster these findings, measurements with more probes should be performed. Research could further be expanded by running the test regime with more TCP ports to determine if TCP blocking is broad or fine tuned to specific ports. A possibility would be that more administrative services, such as SSH for example, experience different on-path filtering.

During the Hurricane Electric incident, mentioned in section 6.3.2, their customer Swisscom did not experience a lot of interruptions. This could indicate some sort of traffic prioritization. The telescope could be able to infer relations between a provider and its different customers by seeing how each customers traffic behaves during high load events of that provider. If the packet loss is not distributed equally between the customers, traffic prioritization is potentially taking place.

Bibliography

- [1] Aditya Akella, Srinivasan Seshan, and Anees Shaikh. “An empirical evaluation of wide-area internet bottlenecks”. In: *Proceedings of the 3rd ACM SIGCOMM Conference on Internet Measurement*. IMC '03. Miami Beach, FL, USA: Association for Computing Machinery, 2003, pp. 101–114. URL: <https://doi.org/10.1145/948205.948219>.
- [2] Manos Antonakakis, Tim April, Michael Bailey, Matt Bernhard, Elie Bursztein, Jaime Cochran, Zakir Durumeric, J. Alex Halderman, Luca Invernizzi, Michalis Kallitsis, Deepak Kumar, Chaz Lever, Zane Ma, Joshua Mason, Damian Menscher, Chad Seaman, Nick Sullivan, Kurt Thomas, and Yi Zhou. “Understanding the Mirai Botnet”. In: *26th USENIX Security Symposium (USENIX Security 17)*. Vancouver, BC: USENIX Association, Aug. 2017, pp. 1093–1110. URL: <https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/antonakakis>.
- [3] AS Rank. <https://doi.org/10.21986/CAIDA.DATA.AS-RANK>.
- [4] Brice Augustin, Xavier Cuvellier, Benjamin Orgogozo, Fabien Viger, Timur Friedman, Matthieu Latapy, Clémence Magnien, and Renata Teixeira. “Avoiding traceroute anomalies with Paris traceroute”. In: Oct. 2006, pp. 153–158.
- [5] Karyn Benson, Alberto Dainotti, KC Claffy, and Emile Aben. “Gaining insight into AS-level outages through analysis of Internet Background Radiation”. In: *2013 IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*. 2013, pp. 447–452.
- [6] John Biggs. *Grum: Inside The Takedown Of One Of The World’s Biggest Spam Networks* | TechCrunch — [techcrunch.com](https://techcrunch.com/2012/08/04/grum-inside-the-takedown-of-one-of-the-worlds-biggest-spam-networks/). <https://techcrunch.com/2012/08/04/grum-inside-the-takedown-of-one-of-the-worlds-biggest-spam-networks/>.
- [7] CAIDA Stardust Documentation. Mar. 2021. URL: <https://www.caida.org/projects/stardust/docs/softwarerepo/>.
- [8] Shinyoung Cho, Romain Fontugne, Kenjiro Cho, Alberto Dainotti, and Phillipa Gill. “BGP hijacking classification”. In: *2019 Network Traffic Measurement and Analysis Conference (TMA)*. 2019, pp. 25–32.
- [9] Cloudflare Learning Center - Memcached DDoS Attacks. URL: <https://www.cloudflare.com/learning/ddos/memcached-ddos-attack/>.
- [10] Alberto Dainotti, Roman Amman, Emile Aben, and Kimberly C. Claffy. “Extracting benefit from harm: using malware pollution to analyze the impact of political and geophysical events on the internet”. In: *SIGCOMM Comput. Commun. Rev.* 42.1 (Jan. 2012), pp. 31–39. URL: <https://doi.org/10.1145/2096149.2096154>.

- [11] Alberto Dainotti, Claudio Squarcella, Emile Aben, Kimberly C. Claffy, Marco Chiesa, Michele Russo, and Antonio Pescapé. “Analysis of Country-Wide Internet Outages Caused by Censorship”. In: *IEEE/ACM Transactions on Networking* 22.6 (2014), pp. 1964–1977.
- [12] Chris Demchak and Yuval Shavitt. “China’s Maxim – Leave No Access Point Unexploited: The Hidden Story of China Telecom’s BGP Hijacking”. In: *Military Cyber Affairs* 3 (June 2018).
- [13] Zakir Durumeric, Eric Wustrow, and J. Alex Halderman. “ZMap: Fast Internet-wide Scanning and Its Security Applications”. In: *22nd USENIX Security Symposium (USENIX Security 13)*. Washington, D.C.: USENIX Association, Aug. 2013, pp. 605–620. URL: <https://www.usenix.org/conference/usenixsecurity13/technical-sessions/paper/durumeric>.
- [14] Claude Fachkha, Elias Bou-Harb, and Mourad Debbabi. “Fingerprinting Internet DNS Amplification DDoS Activities”. In: *2014 6th International Conference on New Technologies, Mobility and Security (NTMS)*. 2014, pp. 1–5.
- [15] *Free Pool of IPv4 Address Space Depleted | The Number Resource Organization — nro.net*. <https://www.nro.net/ipv4-free-pool-depleted>. 2024.
- [16] Oliver Gasser, Quirin Scheitle, Sebastian Gebhard, and Georg Carle. “Scanning the IPv6 Internet: Towards a Comprehensive Hitlist”. In: *Proc. of 8th Int. Workshop on Traffic Monitoring and Analysis*. Louvain-la-Neuve, Belgium, Apr. 2016.
- [17] Chansu Han, Jumpei Shimamura, Takeshi Takahashi, Daisuke Inoue, Masanori Kawakita, Jun’ichi Takeuchi, and Koji Nakao. “Real-Time Detection of Malware Activities by Analyzing Darknet Traffic Using Graphical Lasso”. In: *2019 18th IEEE International Conference On Trust, Security And Privacy In Computing And Communications/13th IEEE International Conference On Big Data Science And Engineering (TrustCom/BigDataSE)*. 2019, pp. 144–151.
- [18] Raphael Hiesgen, Marcin Nawrocki, Alistair King, Alberto Dainotti, Thomas C. Schmidt, and Matthias Wählisch. “Spoki: Unveiling a New Wave of Scanners through a Reactive Network Telescope”. In: *31st USENIX Security Symposium (USENIX Security 22)*. Boston, MA: USENIX Association, Aug. 2022, pp. 431–448. URL: <https://www.usenix.org/conference/usenixsecurity22/presentation/hiesgen>.
- [19] Raphael Hiesgen, Marcin Nawrocki, Thomas C. Schmidt, and Matthias Wählisch. “The Log4j Incident: A Comprehensive Measurement Study of a Critical Vulnerability”. In: *IEEE Transactions on Network and Service Management* (2024), pp. 1–1.
- [20] Thomas Holterbach, Thomas Alfroy, Amreesh Phokeer, Alberto Dainotti, and Cristel Pelsser. “A System to Detect Forged-Origin BGP Hijacks”. In: *21st USENIX Symposium on Networked Systems Design and Implementation (NSDI 24)*. Santa Clara, CA: USENIX Association, Apr. 2024, pp. 1751–1770. URL: <https://www.usenix.org/conference/nsdi24/presentation/holterbach>.
- [21] Lee Howard and Peter Tobey. *IPv4 address prices*. <https://ipv4.global/blog/ipv4-address-prices/>. June 2023.
- [22] *Internet Outage Detection and Analysis (IODA)*. Oct. 2023. URL: <https://www.caida.org/projects/ioda/>.
- [23] Peter Kacherginsky. *Celer Bridge incident analysis*. <https://www.coinbase.com/de/blog/celer-bridge-incident-analysis>. Sept. 2022.
- [24] Nick Kephart. *Route leak causes global outage in level 3 network*. June 2015. URL: <https://www.thousandeyes.com/blog/route-leak-causes-global-outage-level-3-network>.

- [25] Johannes Krupp, Michael Backes, and Christian Rossow. “Identifying the Scan and Attack Infrastructures Behind Amplification DDoS Attacks”. In: *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*. CCS ’16. Vienna, Austria: Association for Computing Machinery, 2016, pp. 1426–1437. URL: <https://doi.org/10.1145/2976749.2978293>.
- [26] Matt Lepinski and Stephen Kent. *An Infrastructure to Support Secure Internet Routing*. RFC 6480. Feb. 2012. URL: <https://www.rfc-editor.org/info/rfc6480>.
- [27] Philipp M. Lutscher, Nils B. Weidmann, Margaret E. Roberts, Mattijs Jonker, Alistair King, and Alberto Dainotti. “At Home and Abroad: The Use of Denial-of-service Attacks during Elections in Nondemocratic Regimes”. In: *Journal of Conflict Resolution* 64.2-3 (2020), pp. 373–401. eprint: <https://doi.org/10.1177/0022002719861676>. URL: <https://doi.org/10.1177/0022002719861676>.
- [28] Doug Madory and Job Snijders. *Exploring the latest RPKI ROV adoption numbers*. May 2023. URL: <https://www.kentik.com/blog/exploring-the-latest-rpki-rov-adoption-numbers/>.
- [29] Athina Markopoulou, Fouad Tobagi, and Mansour Karam. “Loss and Delay Measurements of Internet Backbones”. In: *Computer Communications* 29.10 (2006), pp. 1590–1604. URL: <https://www.sciencedirect.com/science/article/pii/S0140366405002720>.
- [30] Prodosh Mohapatra, John Scudder, David Ward, Randy Bush, and Rob Austein. *BGP Prefix Origin Validation*. RFC 6811. Jan. 2013. URL: <https://www.rfc-editor.org/info/rfc6811>.
- [31] Robert Moskowitz, Daniel Karrenberg, Yakov Rekhter, Eliot Lear, and Geert Jan de Groot. *Address Allocation for Private Internets*. RFC 1918. Feb. 1996. URL: <https://www.rfc-editor.org/info/rfc1918>.
- [32] Marcin Nawrocki, Raphael Hiesgen, Thomas C. Schmidt, and Matthias Wählisch. “QUICsand: quantifying QUIC reconnaissance scans and DoS flooding events”. In: *Proceedings of the 21st ACM Internet Measurement Conference*. IMC ’21. Virtual Event: Association for Computing Machinery, 2021, pp. 283–291. URL: <https://doi.org/10.1145/3487552.3487840>.
- [33] Marcin Nawrocki, Mattijs Jonker, Thomas C. Schmidt, and Matthias Wählisch. “The far side of DNS amplification: tracing the DDoS attack ecosystem from the internet core”. In: *Proceedings of the 21st ACM Internet Measurement Conference*. IMC ’21. Virtual Event: Association for Computing Machinery, 2021, pp. 419–434. URL: <https://doi.org/10.1145/3487552.3487835>.
- [34] Marcin Nawrocki, Maynard Koch, Thomas C. Schmidt, and Matthias Wählisch. “Transparent forwarders: an unnoticed component of the open DNS infrastructure”. In: *Proceedings of the 17th International Conference on Emerging Networking EXperiments and Technologies*. CoNEXT ’21. Virtual Event, Germany: Association for Computing Machinery, 2021, pp. 454–462. URL: <https://doi.org/10.1145/3485983.3494872>.
- [35] RIPE NCC. *YouTube hijacking: A ripe NCC RIS case study*. Mar. 2008. URL: <https://www.ripe.net/publications/news/youtube-hijacking-a-ripe-ncc-ris-case-study/>.
- [36] William B. Norton. *The art of peering: The peering playbook*. URL: <https://drpeering.net/white-papers/Art-Of-Peering-The-Peering-Playbook.html>.
- [37] Ramakrishna Padmanabhan, Aaron Schulman, Dave Levin, and Neil Spring. “Residential links under the weather”. In: *Proceedings of the ACM Special Interest Group on Data Communication*. SIGCOMM ’19. Beijing, China: Association for Computing Machinery, 2019, pp. 145–158. URL: <https://doi.org/10.1145/3341302.3342084>.

- [38] Eric Pauley, Paul Barford, and Patrick McDaniel. “DScope: A Cloud-Native Internet Telescope”. In: *32nd USENIX Security Symposium (USENIX Security 23)*. Anaheim, CA: USENIX Association, Aug. 2023, pp. 5989–6006. URL: <https://www.usenix.org/conference/usenixsecurity23/presentation/pauley>.
- [39] Eric Pauley, Ryan Sheatsley, Blaine Hoak, Quinn Burke, Yohan Beugin, and Patrick McDaniel. “Measuring and Mitigating the Risk of IP Reuse on Public Clouds”. In: *2022 IEEE Symposium on Security and Privacy (SP)*. IEEE, May 2022. URL: <http://dx.doi.org/10.1109/SP46214.2022.9833784>.
- [40] *PeeringDB - AS3356*. 2024. URL: <https://www.peeringdb.com/net/504>.
- [41] Louis Poinsignon, Dina Kozlov, Suleman Ahmad, Jonathan Hoyland, and Sudheesh Singanamalla. *BGP leaks and cryptocurrencies*. Oct. 2023. URL: <https://blog.cloudflare.com/bgp-leaks-and-crypto-currencies/>.
- [42] Yakov Rekhter and Tony Li. *A Border Gateway Protocol 4 (BGP-4)*. RFC 1654. July 1994. URL: <https://www.rfc-editor.org/info/rfc1654>.
- [43] Johann Schlamp, Ralph Holz, Quentin Jacquemart, Georg Carle, and Ernst W. Biersack. “HEAP: Reliable Assessment of BGP Hijacking Attacks”. In: *IEEE Journal on Selected Areas in Communications* 34.6 (2016), pp. 1849–1861.
- [44] Aftab Siddiqui. *KlaySwap - Another BGP Hijack Targeting Crypto Wallets - MANRS* — [manrs.org](https://manrs.org/2022/02/klayswap-another-bgp-hijack-targeting-crypto-wallets/). <https://manrs.org/2022/02/klayswap-another-bgp-hijack-targeting-crypto-wallets/>. Feb. 2022.
- [45] Aftab Siddiqui. *Major route leak by AS28548 - another BGP optimizer?* Feb. 2021. URL: <https://manrs.org/2021/02/major-route-leak-by-as28548-another-bgp-optimizer/>.
- [46] Raffaele Sommesse, KC Claffy, Roland van Rijswijk-Deij, Arnab Chattopadhyay, Alberto Dainotti, Anna Sperotto, and Mattijs Jonker. “Investigating the impact of DDoS attacks on DNS infrastructure”. In: *Proceedings of the 22nd ACM Internet Measurement Conference*. IMC ’22. Nice, France: Association for Computing Machinery, 2022, pp. 51–64. URL: <https://doi.org/10.1145/3517745.3561458>.
- [47] *Spoofers Project - CAIDA*. URL: https://spoofers.caida.org/as_stats.php.
- [48] *UPD scans - nmap*. URL: <https://nmap.org/book/scan-methods-udp-scan.html>.
- [49] Pierre-Antoine Vervier, Quentin Jacquemart, Johann Schlamp, Olivier Thonnard, Georg Carle, Guillaume Urvoy-Keller, Ernst Biersack, and Marc Dacier. “Malicious BGP hijacks: Appearances can be deceiving”. In: *2014 IEEE International Conference on Communications (ICC)*. 2014, pp. 884–889.
- [50] Daniel Wagner, Sahil Ashish Ranadive, Harm Griffioen, Michalis Kallitsis, Alberto Dainotti, Georgios Smaragdakis, and Anja Feldmann. “How to Operate a Meta-Telescope in your Spare Time”. In: *Proceedings of the 2023 ACM on Internet Measurement Conference*. IMC ’23. Montreal, Canada: Association for Computing Machinery, 2023, pp. 328–343. URL: <https://doi.org/10.1145/3618257.3624831>.
- [51] Gerry Wan, Liz Izhikevich, David Adrian, Katsunari Yoshioka, Ralph Holz, Christian Rossow, and Zakir Durumeric. “On the Origin of Scanning: The Impact of Location on Internet-Wide Scans”. In: *Proceedings of the ACM Internet Measurement Conference*. IMC ’20. Virtual Event, USA: Association for Computing Machinery, 2020, pp. 662–679. URL: <https://doi.org/10.1145/3419394.3424214>.
- [52] Guowu Xie, Huy Hang, and Michalis Faloutsos. “Scanner hunter: understanding HTTP scanning traffic”. In: *Proceedings of the 9th ACM Symposium on Information, Computer and Communications Security*. ASIA CCS ’14. Kyoto, Japan: Association for Computing Machinery, 2014, pp. 27–38. URL: <https://doi.org/10.1145/2590296.2590297>.